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Master Thesis

Handling Missing Data in Multilevel Models with Few Clusters: A Simulation Study

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Abstract

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Multilevel studies often involve two challenges simultaneously: small numbers of clusters and missing data. However, the performance of modern missing-data methods under these combined conditions remains insufficiently understood, particularly in models with missing predictor variables and contextual effects. This thesis examines the performance of multiple imputation, using both conventional and small-sample-corrected degrees of freedom, and compares it with a fully Bayesian approach in a simulation study. The study features a simple random-intercept model with a manifest group-mean predictor, with missings in the predictor variable and one level-2 auxiliary variable. Simulation conditions varied in terms of intraclass correlation (.1 vs. .3), missingness mechanism (MAR vs. MCAR), between-group effect size (0 vs. 0.4), and level-2 sample size (15, 30, 60). Method performance was evaluated in terms of bias, coverage, standard deviation of point estimates, and interval width. Results indicate a trade-off between bias and interval width for the between-group effect: multiple imputation tended to produce more biased estimates, particularly under low intraclass correlation conditions, whereas the Bayesian approach yielded wider credible intervals. As coverage under conventional degrees of freedom was already close to the nominal level, the small-sample correction tended to result in overcoverage. The findings are discussed with regard to methodological limitations and their implications for applied multilevel research.

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Chapter 1

Introduction

Multilevel models are a standard statistical tool for analyzing clustered data, which frequently occur in the social sciences (e.g., patients nested within therapists, students within classes, or employees within firms). In practice, such data sets are often characterized by a relatively small number of groups and by the presence of missing values (Maas & Hox, 2005; McNeish & Stapleton, 2016b; Nicholson et al., 2017; Peugh & Enders, 2004).

Methodological research has proposed a range of approaches to address each of these challenges separately. Small-sample corrections for multilevel models have been developed to improve statistical inference when the number of clusters is limited (Elff et al., 2021; Holtmann et al., 2016; McNeish & Stapleton, 2016b, 2016a). At the same time, modern missing-data methods have been extended to multilevel data structures (see e.g. Audigier et al., 2018; Enders et al., 2016; Enders, 2025; Grund et al., 2019; Kahan et al., 2016; Yucel, 2008).

Some research has investigated how these two challenges interact, both for single-level regressions (see e.g. McNeish, 2017a; Von Hippel, 2013) and in multilevel contexts (e.g., McNeish, 2018; Shin et al., 2025; Shin & Shin, 2025; Zitzmann et al., 2015). However, the existing literature remains limited, particularly for multilevel models in which effects of level-2 variables are of substantive interest. Such level-2 variables may be directly measured group characteristics (e.g., class size) or level-1 variables that are included to estimate contextual effects, such as the mean cognitive ability of students in a class (Croon & Van Veldhoven, 2007; Lüdtke et al., 2008; Raudenbush & Bryk, 2010; Snijders & Bosker, 2012). Contextual models are widely used in psychological research (e.g., Anderman, 2002; Papaioannou et al., 2004), but missing data become especially relevant in this setting. Incomplete predictor variables generally pose additional challenges for statistical inference compared to situations in which only the outcome variable is incomplete (Bartlett et al., 2015; Enders, 2022). Additionally, in models with manifest group means, previous work suggests that likelihood-based approaches to incomplete predictors may produce biased estimates of between-group effects (Grund et al., 2019), which can make alternative approaches such as fully Bayesian estimation and multiple imputation more appealing in such settings.

This thesis therefore investigates the suitability of modern approaches to handling missing data in small-sample multilevel models with contextual effects in which predictor variables are partially missing. Specifically, the performance of multiple imputation and fully Bayesian estimation is evaluated. The structure of this thesis is as follows: First, the theoretical foundations of multilevel models are introduced, and current methodological approaches for handling small samples and missing data are reviewed. Subsequently, a simulation study is conducted to investigate the performance of these methods under different data conditions. Finally, the findings are discussed with regard to their implications for handling missing predictor data in small-sample multilevel models, and the limitations of the study are considered.

Chapter 2

Theory

2.1 Theoretical foundation of multilevel models

Populations in which units of interest (e.g., students, employees, or musicians) are grouped into higher-level units such as classes, firms, or ensembles, are common in the social sciences. Imagine, for example, that you are interested in the musical self-concept of young adults and whether it is related to how many hours they practice their instrument each week. To investigate this question, you collect data from several youth ensembles. You recruit 15 ensembles, each consisting of 10 players. Every player completes a questionnaire assessing their musical self-concept as well as their weekly practice time. At first glance, this seems like a simple data set of 150 individuals. At closer inspection, however, players from the same ensemble rehearse together, share teachers, and experience similar social and musical environments. Their responses may therefore be more similar within ensembles than between ensembles.

Data of this kind are called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7). Such data for example arises from a multistage sampling process, in which groups are sampled first and individuals are subsequently sampled within these groups[1]. This process induces dependencies in the data, as observations within the same group are not selected independently (Snijders & Bosker, 2012, p. 7). Beyond the sampling process, individuals within a group typically share contextual influences (e.g., the same ensemble teacher or classroom environment), which leads to increased similarity in their responses compared to individuals from another group. As a consequence, the assumption of independent errors in traditional linear models is violated, because residuals within groups are correlated (Raudenbush & Bryk, 2010). A widely used approach to account for this structure is the use of *multilevel models* (also referred to as *hierarchical linear models* or *mixed models*), which are introduced in the following section.

2.1.1 The concept of multilevel regression models

The aim of regression analysis, whether single- or multilevel, is to explain variance in an outcome variable by including associated variables in the model as predictors (Snijders & Bosker, 2012, p. 80). In single-level regression, the residual e_i captures the unexplained variance in Y, that is differences in the outcome from the prediction that the model made (Leonhart, 2013, p. 319). To account for the dependency of the observations within groups, multilevel models extend this regular linear regression by adding level-2 random effects to explain the variability between groups that cannot

[1] In this thesis, the terms 'groups' or 'clusters' refer to higher-level units, while lower-level units are referred to as 'persons', 'individuals' or 'observations'. These models can also be applied to longitudinal data, where measurement occasions are nested within individuals, and to data structures with more than two levels.

be accounted for by predictors (Snijders & Bosker, 2012, p. 80). This partition of variance can be expressed in the multilevel regression model without predictors, the *empty model*:

$$Y_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

Y_{ij} is the outcome value of person i in group j, for $i = 1, ..., n_j$ (n_j = number of observations in the group) and $j = 1, ..., N_2$ (N_2 = number of groups). This outcome is modeled as the sum of a fixed intercept parameter γ_{00}, a group-level random effect u_{0j} and an individual-level residual, e_{ij} (Snijders & Bosker, 2012, p. 49). The fixed intercept parameter γ_{00} can be described as the overall mean in outcome Y_{ij} across persons and groups. The random intercept u_{0j} induces correlation in the outcome value between individuals within the same group, as all individuals in group j have the same u_{0j} value. Conditional on this group component, however, the remaining level-1 residuals e_{ij} are assumed to be independent. In this way, the model explicitly accounts for the dependency of observations that would that would violate the assumptions of a single-level regression model. In Figure 2.1, the empty model is illustrated on the left side.

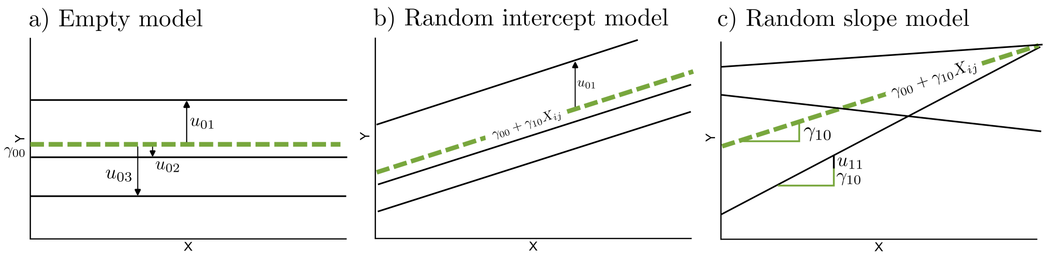

FIGURE 2.1: Schematic illustration of common multilevel model specifications in the level-1 coordinate space: empty model, random intercept model, and random slope model. The green line represents the fixed-effect prediction averaged across groups. The black lines represent exemplary groups and their deviations from the fixed effects. Individual-level observations are not displayed.

2.1.2 Level-1 and level-2 predictors in multilevel regression models

Predictors can then be included to explain variability at level 1 and level 2. At level 1, the outcome of a person i in group j is modeled as a function of individual-level explanatory variables X_{ij}. For simplicity, the level-1 equation is given with a single level-1 predictor:

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + e_{ij} \tag{2.1}$$

In this equation, β_{0j} is the intercept for group j, and β_{1j} is the corresponding slope in group j. In a random slope model, each group is allowed to have its own intercept and slope. This model is illustrated on the right in Figure 2.1. The variation between groups in these coefficients is modeled at level 2, where β_{0j} and β_{1j} can be decomposed into *fixed effects*, representing the average coefficients across groups (γ_{00} for the intercept and γ_{10} for the slope), and group-specific *random effects*, representing deviations from these averages (u_{0j} and u_{1j}, respectively; Snijders & Bosker (2012), p. 75). Without level-2 predictors, the level-2 equations are:

$$\beta_{0j} = \underbrace{\gamma_{00}}_{\text{Fixed part}} + \underbrace{u_{0j}}_{\text{Random part}}$$

$$\beta_{1j} = \underbrace{\gamma_{10}}_{\text{Fixed part}} + \underbrace{u_{1j}}_{\text{Random part}}$$

In order to explain between-group variability in the intercepts and slopes, level-2 predictors Z_j may be included (Snijders & Bosker, 2012, p. 80). Including such predictors may reduce the unexplained variance of the random effects u_{0j} and u_{1j}. With a single level-2 predictor, the model becomes:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j} \tag{2.2}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} Z_j + u_{1j} \tag{2.3}$$

Here, the level-2 predictor Z_j is used to explain systematic between-group differences in the level-1 intercepts and slopes. The coefficient γ_{01} represents the expected difference in the group-specific intercept β_{0j} associated with a one-unit difference in Z_j. Similarly, γ_{11} represents the expected difference in the group-specific slope β_{1j} associated with a one-unit difference in Z_j. Thus, γ_{11} describes a cross-level interaction: it indicates whether the association between the individual-level predictor X_{ij} and the outcome Y_{ij} differs depending on the group-level variable Z_j. The residual terms u_{0j} and u_{1j} represent remaining between-group variation in intercepts and slopes that is not explained by Z_j. Substituting the level-2 equations Equation 2.2 and Equation 2.3 into the level-1 equation Equation 2.1 yields the combined multilevel model:

$$Y_{ij} = \underbrace{\gamma_{00} + \gamma_{10} X_{ij} + \gamma_{01} Z_j + \gamma_{11} X_{ij} Z_j}_{\text{Fixed part}} + \underbrace{u_{0j} + u_{1j} X_{ij} + e_{ij}}_{\text{Random part}}$$

In many empirical applications of multilevel models, the slope β_{1j} is assumed to be constant across groups, so that no random slope component is included ($u_{1j} = 0$). This leads to a *random-intercept model* (Snijders & Bosker, 2012, p. 45). In its simpler form, the model also does not include a cross-level interaction ($\gamma_{11} = 0$). [2]. This model allows for baseline differences in the outcome between groups, whereas the strength of the relationship between the level-1 predictor and the outcome is assumed to be constant across groups. This model is illustrated in the middle panel in Figure 2.1. The corresponding level-2 equations are:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

The combined random-intercept equation therefore has only two random components, the group-level random effect u_{0j} and the individual-level residual e_{ij}:

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + \gamma_{01} Z_j + u_{0j} + e_{ij}$$

Distributional assumptions of the multilevel regression model Similar distributional assumptions are typically made for the level-2 random effects as for the residuals in single-level regression: they have mean zero, constant variance across clusters, and are independent of each other across clusters and independent of the level-1 residuals.

[2] A cross-level interaction can technically be included in a random-intercept model. However, this assumes that slope differences are modeled only as a systematic function of the level-2 predictor, without allowing for remaining unexplained slope variation. For this reason, cross-level interactions are usually discussed together with random slopes; see Heisig & Schaeffer (2019); Aguinis et al. (2013).

However, random effects within the same cluster - i.e. the random intercept u_{0j} and random slope u_{1j} - may be correlated and are often expected to do so (Snijders & Bosker, 2012, p. 76). Additionally, random effects are commonly assumed to follow a normal distribution for convenience, as it facilitates the likelihood-based estimation of the unknown parameters (Fahrmeir et al., 2021, p. 388). For a model with both a random intercept and a random slope, these assumptions can be summarized by a bivariate normal distribution,

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{T}\right), \qquad \mathbf{T} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix} \quad ,$$

where τ_0^2 denotes the variance of the random intercepts across groups, and τ_1^2 denotes the variance of the random slopes across groups. The covariance τ_{01} captures the association between random intercepts and random slopes within the same group. For example, a positive covariance would indicate that groups with higher-than-average intercepts also tend to have steeper-than-average slopes.

The level-1 residuals are typically assumed to be normally distributed with mean zero and constant variance (σ^2):

$$e_{ij} \sim \mathcal{N}(0, \sigma^2)$$

Furthermore they are assumed to be independent of the level-2 random effects:

$$\mathrm{Cov}(u_{0j}, e_{ij}) = 0, \qquad \mathrm{Cov}(u_{1j}, e_{ij}) = 0.$$

2.1.3 Contextual effects in multilevel models

As mentioned previously, individuals nested within the same group share a common context. In some cases, variables measured at the individual level may therefore be relevant in two different ways. First, they may describe how individuals differ from other individuals within the same group. Second, when aggregated to the group level, they may describe characteristics of the group context itself. Consider again the ensemble example, where students' practice duration was used to explain the musical self concept of students. A student who practices more than their peers may feel more competent and report a higher musical self-concept. But at the same time, students often compare their own practice behavior with that of other ensemble members; and students in ensembles where everyone practices a lot may feel less competent, because upward social comparisons become more likely. This leads to two possible hypotheses:

1. Students who practice more than others in their ensemble tend to report a higher musical self-concept.
2. Students in ensembles with a higher average practice time may report a lower musical self-concept.

   This example illustrates why it can be important to distinguish within-group and between-group associations. The within-group association describes how individual deviations from the group mean are related to the outcome. The between-group association describes how differences between group means are related to the outcome. A contextual effect is present when the group-level aggregate is associated with the outcome after controlling for individual differences within groups (Raudenbush & Bryk, 2010, p. 139). A random-intercept model provides a convenient framework for such analyses. To disentangle these processes, group means of a level-1 predictor may be included in the model as additional level-2 predictors. To separate within- and between-group effects, the level-1 predictor is commonly group-mean-centered: Instead of using X_{ij} directly, the centered predictor $(X_{ij} - \bar{X}_j)$ is entered into the model (Snijders & Bosker, 2012, p. 58). A random intercept model including such a decomposition can be written like this[3]:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_j) + \gamma_{01}\bar{X}_j + u_{0j} + e_{ij} \tag{2.4}$$

In this model, the effect of weekly practice duration on Y_{ij}, the musical self-concept of student i in ensemble j, is separated into the *within-group effect* γ_{10}, indicating how a student's own practice time relates to their musical self-concept while controlling for the ensemble context, and the *between-group effect* γ_{01}, indicating how the ensemble's average practice time relates to students' musical self-concept, controlling for individual deviations from the group mean. The difference between these two coefficients, γ_{01}-γ_{10}, represents the *contextual effect* (Raudenbush & Bryk, 2010, p. 139). For the remainder of this thesis, the focus lies on this particular random intercept model.

2.1.4 The intraclass correlation coefficient (ICC)

In the random intercept model, the variance components are given by

$$u_{0j} \sim \mathcal{N}(0, \tau_0^2), \qquad e_{ij} \sim \mathcal{N}(0, \sigma^2), \qquad \mathrm{Cov}(e_{ij}, u_{0j}) = 0.$$

From these variance components, the proportion of variability that is attributable to between-group differences can be calculated. This quantity is referred to as the *intraclass correlation coefficient* (ICC). In the empty model, the ICC is equal to the correlation between the outcomes of two randomly chosen individuals from the same group (Snijders & Bosker, 2012, p. 18). In models including predictors, the ICC is interpreted conditionally on the predictors, that is, as the correlation between two individuals from the same group who share identical predictor values (also referred to as the *residual ICC*) (Snijders & Bosker, 2012, p. 52). Formally, the ICC in a random intercept model is defined as

$$ICC = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}$$

The ICC is an important diagnostic metric because it indicates the extent to which residual variance in Y_{ij} is attributable to differences between groups. If ICC $= 0$, no residual between-group variance is estimated in the model. Larger ICC values, in contrast, indicate increasing similarity of observations within groups and hence stronger statistical dependency. Ignoring this dependency and applying a classical linear model can lead to biased standard errors, confidence intervals, and hypothesis tests (Aarts et al., 2014; Fahrmeir et al., 2021, p. 370).

2.1.5 Estimation of parameters in multilevel models

The random intercept model is defined by the fixed effects γ and the variance components of the random effects, σ^2 and τ_0^2 (Snijders & Bosker, 2012, p. 60). In practice, however, these model parameters are unknown and must be estimated from the observed data. In many applications, including the contextual-effect setting considered in this thesis, substantive hypotheses primarily concern fixed effects. In

[3]The group mean $\bar{X}_j$ is treated as an observed (manifest) aggregate in this approach. Group means can also be modeled as latent variables, thereby accounting for measurement error and sampling variability (Lüdtke et al., 2008).

the ensemble example introduced above, for instance, the within- and between-group associations between practice time and musical self-concept would be represented by fixed-effect parameters. The variance components are nevertheless essential, because they influence the estimated uncertainty of the fixed-effect estimates. Based on the resulting parameter estimates, their estimated uncertainty, and assumptions about the sampling distribution of the estimators, inferences about the corresponding population parameters are drawn (Raudenbush & Bryk, 2010, p. 57f.). The steps of this process are illustrated in more detail in Figure 2.2. Aspects of the observed data may pose challenges to the reliable performance of this 'inferential machinery'. Two central challenging aspects, namely small sample sizes and missing data, will be discussed in the following chapters.

2.2 Small sample sizes in multilevel models

Frequentist estimation procedures for linear multilevel models typically rely on large-sample (asymptotic) theory. In multilevel models, the number of groups is particularly important. Consider again the musical self-concept study with 150 students nested in 15 ensembles. Although 150 participants may seem like a reasonably large sample, the number of ensembles is still small. Information about between-ensemble variation and level-2 effects is therefore based on only 15 higher-level units. Simulation studies suggest that approximately 25–30 clusters are often required for reliable estimation, depending on the model specification and the parameters of interest (Maas & Hox, 2005; McNeish, 2017a; McNeish & Stapleton, 2016b), but this is often difficult to achieve because recruiting a large number of groups can be challenging and costly. When the number of groups is small, several components of the inferential procedure may perform poorly if no corrections are used, particularly the estimation of level-2 variance components, the calculation of standard errors, and the approximation of degrees of freedom for fixed-effect tests (Elff et al., 2021; Maas & Hox, 2005; McNeish, 2017b).

Estimation of coefficients and variance components. For correctly specified linear multilevel models, ML estimates of fixed effect coefficients are unbiased under quite general conditions (Elff et al., 2021, Appendix A.3; Kackar & Harville, 1981). Importantly, this unbiasedness result does not depend on the level-1 or level-2 sample size and a small number of groups therefore does not in itself imply biased fixed-effect point estimates. The estimation of variance components, however, is more delicate. In ML estimation, fixed effects and variance components are estimated from the same likelihood, typically via iterative procedures like the Expectation Maximization (EM) algorithm (Raudenbush & Bryk, 2010, p. 52). This leads to systematic underestimation of variance components in small samples, because the uncertainty about the fixed effects is not accounted for (Raudenbush & Bryk, 2010, p. 53). REML addresses this issue by separating the estimation conceptually, which leads to less downward-biased variance component estimates, while ML and REML become asymptotically equivalent as sample size increases (McNeish, 2017b). Nevertheless, even under REML, the estimated variance of the fixed-effect estimators may still be too small in small samples for two reasons. First, the variance formula treats the estimated variance components $\widehat{\theta}$ as known and therefore ignores their sampling variability (Kackar & Harville, 1984). Second, the REML variance expression itself relies on asymptotic approximations (McNeish, 2017b). Both issues can lead to underestimated standard errors. Several small-sample corrections have been proposed to address this additional problems, among which the Kenward–Roger procedure is

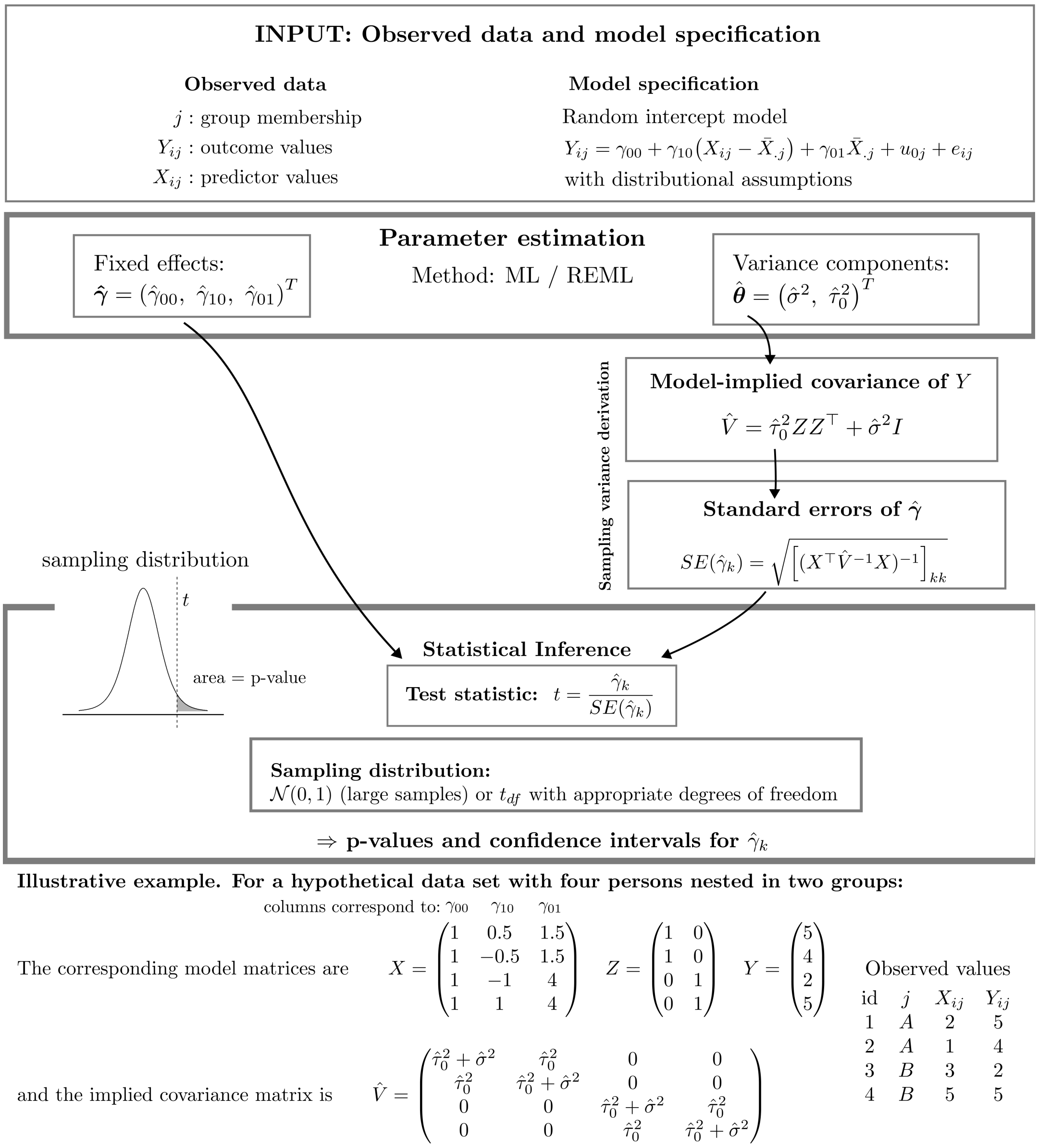

FIGURE 2.2: Schematic representation of the frequentist inferential process in a random intercept model, based on Snijders & Bosker (2012) and Raudenbush & Bryk (2010). In the first step, fixed effects and variance components are estimated from the observed data, commonly using maximum likelihood (ML) or restricted maximum likelihood (REML) estimation. The estimated random-intercept variance $\hat{\tau}_0^2$ and residual variance $\hat{\sigma}^2$, together with the random-effects design matrix Z, determine the estimated variance-covariance matrix of the outcome vector, $\hat{V}$. In the random-intercept model, Z encodes group membership and thereby represents that observations from the same group share the same random intercept. The estimated uncertainty of the fixed-effect estimates is then derived from $\hat{V}$ and the fixed-effects design matrix X. This matrix contains one row for each observation and columns for the fixed-effect terms in the model. For a given fixed effect $\hat{\gamma}_k$, the corresponding standard error quantifies the estimated sampling variability of the estimator. The point estimate and its standard error are used to form a test statistic, which is evaluated against an assumed sampling distribution to obtain p-values and confidence intervals. Thus, frequentist inference for fixed effects depends not only on the point estimates themselves, but also on the estimated variance components and on assumptions about the sampling distribution of the estimators.

one of the most widely used (McNeish, 2017b).

Choice of degrees of freedom. REML estimation and small-sample adjustments such as Kenward–Roger can reduce some of the inferential problems described above. However, inference also requires choosing an appropriate reference distribution for test statistics and confidence intervals. In small samples, a t-distribution is typically recommended instead of a standard normal distribution (Snijders & Bosker, 2012, p. 95). Its shape is determined by the degrees of freedom, which can be understood as the amount of independent information remaining after accounting for estimated parameters (Pandey & Bright, 2008). In multilevel models, degrees of freedom generally cannot be calculated exactly and must therefore be approximated (Snijders & Bosker, 2012, p. 95). Different approximations are available. For example, the Kenward–Roger procedure includes a small-sample approximation of the degrees of freedom based on the Satterthwaite method (McNeish, 2017b). A simpler approximation, described by Raudenbush & Bryk (2010, p. 58), Snijders & Bosker (2012, p. 95), and Elff et al. (2021), follows the general idea of degrees of freedom as information minus estimated parameters. For level-1 coefficients, the degrees of freedom are approximated by the total number of level-1 observations minus the number of level-1 predictors and the intercept. For level-2 coefficients, the degrees of freedom are approximated by the number of level-2 units minus the number of level-2 predictors and the intercept:

$$df_{Lv1} = N_1 - r - 1, \qquad df_{Lv2} = N_2 - q - 1 \tag{2.5}$$

where N_1 is the total number of level-1 observations, N_2 is the number of level-2 units, r is the total number of predictors, and q is the number of level-2 predictors.

Bayesian estimation as an alternative. As an alternative to the frequentist framework predominantly presented here, Bayesian estimation is often recommended for small samples (Depaoli & Clifton, 2015; Kadane, 2015; McNeish, 2016b; Van De Schoot et al., 2014). Unlike frequentist inference based on asymptotic sampling distributions, Bayesian inference represents uncertainty through posterior distributions and therefore does not rely on small-sample corrections for standard errors or degrees of freedom (McNeish, 2016b). The incorporation of prior information about parameters may further improve estimation accuracy when the data provide limited information (Gelman et al., 2025, p. 355; Smid et al., 2020). Bayesian estimation will be revisited below in the context of missing data.

2.3 Missing data in multilevel models

In the study about musical self-concept, you could find after data collection that some students in several ensembles did not report how many hours they practice per week. They completed the questionnaire on musical self-concept but left the practice question blank. The reason for this is unknown: they might have forgotten to answer the question, skipped it intentionally, or misunderstood it. Regardless of the reason, the data set now contains missing values. How should these missing observations be handled in the analysis?

Missing data (i.e., incomplete data sets) are a widespread issue in empirical research and of course also apply to multilevel data, for example, because individuals drop out of studies or forget to fill out certain items of a questionnaire. Three major concerns commonly arise from data that contain missing values (Lüdtke et al., 2007):

1. The effective sample size is reduced, leading to lower efficiency in estimation (this is especially troublesome if the would-be sample size is already small)

2. Typical statistical methods are difficult to conduct, because they were designed for complete data.
3. Parameter estimation may be biased, if the missing and the observed data differ from each other systematically

Therefore, it is crucial to treat missing data appropriately. In the last three decades, this issue has received attention in psychological research, and modern methods were developed (Enders, 2022; Schafer & Graham, 2002), more recently also for the more complex case of multilevel data (Enders, 2025; Lüdtke et al., 2017). This chapter discusses these methods, after giving a short introduction into different missing data mechanisms and patterns.

2.3.1 Missing data mechanisms

Missing values may arise for different reasons, and different missing-data mechanisms can have different consequences for statistical inference. Rubin (1976) introduced the formal framework for these mechanisms, which was later systematized in Little and Rubin's textbook (2020). These mechanisms also apply to multilevel data, but the situation becomes more complex because missing values can occur in the outcome, in level-1 predictors, in level-2 predictors, in the grouping variable, or in combinations of these variables (Van Buuren, 2018). The missing-data mechanism is generally not empirically testable from the observed data alone; assumptions about the mechanism must therefore be theoretically justified (Enders, 2022, pp. 14–17).

Missing completely at random (MCAR). The hypothetical complete data matrix Y_{com} can be partitioned into observed, Y_{obs}, and missing parts, Y_{mis}. R denotes the missingness[4]. Missing values are considered to be MCAR if the distribution of missingness is independent of both the observed and the unobserved data (Schafer & Graham, 2002):

$$P(R \mid Y_{com}) = P(R)$$

In other words, no variable (observed or unobserved) is predictive of missingness and every individual has the same chance of missing values (Enders, 2022, p. 6).

Missing at random (MAR). The MAR mechanism describes the case when the probability of missingness may depend on observed data, but not on the unobserved values themselves once the observed data are taken into account (Enders, 2022, p. 8):

$$P(R \mid Y_{com}) = P(R \mid Y_{obs})$$

Missingness is therefore not random, as the name would suggest, but *conditionally random*: After controlling for the observed data, the missingness is purely random (Enders, 2022, p. 8). For example, younger students may be more likely to skip the question on weekly practice duration, for instance due to reduced concentration toward the end of the questionnaire. If age is fully observed and included in the missing-data procedure, the MAR mechanism may be plausible.

Missing not at random (MNAR). If missingness depends on unobserved values, this is called a missing not at random process:

$$P(R \mid Y_{com}) = P(R \mid Y_{obs}, Y_{mis})$$

Two individuals with identical scores in all observed variables therefore no longer have the same chance of missing values, because this chance depends on the unavailable,

[4] R is treated as a set of random variables that indicate for each value if it is missing or not (Schafer & Graham, 2002).

missing information itself (Enders, 2022, p. 11). For example, students who practice less may be more likely to skip the question on weekly practice duration. If no observed variables can account for this behavior, missingness depends on the practice duration itself, which represents a MNAR mechanism.

The approaches mentioned in this thesis generally rely on the assumption that data are MAR. Methods for dealing with MNAR processes are not discussed in this thesis.

2.3.2 Methods for handling missing data

Many estimation procedures require complete data. Therefore, missing data must be handled explicitly before model estimation. If researchers do not actively specify a method, statistical software often defaults to using only complete observations (Lüdtke et al., 2007). This simple ad hoc procedure is called *listwise deletion* (LD) or *complete case analysis*. Cases with at least one missing value are discarded, and the model is estimated using only complete observations (Graham, 2009). In multilevel settings, missingness on the group level implies that all individuals in this group are excluded as well.

LD is generally not recommended, because parameter estimates can be substantially biased when missingness is not MCAR. In addition, partial data is unused, which reduces statistical power (Enders, 2022, p. 24). More modern and better-performing methods are available and will be introduced in the following sections. These approaches can be broadly divided into two classes:
1. *Model-based approaches* to missing data base parameter estimation and inference directly on the incomplete data, for example via the observed-data likelihood or the posterior distribution (Little & Rubin, 2020, p. 25). 2. *Imputation-based approaches* first replace missing values by one or more plausible values, creating one or more completed data sets that can then be analyzed using standard complete-data methods.

2.3.2.1 Multiple Imputation

Multiple imputation (MI) is an imputation-based method, as the name suggests. The basic idea is to replace missing values with plausible draws (“imputations”) from their predictive distribution. The substantive analysis model is then fitted to the imputed data. In order to include the uncertainty due to missing values, in MI several imputations for each missing observation are generated. By incorporating the variance between these imputations, the additional uncertainty can be captured (Enders, 2022, p. 283). The analysis of incomplete data is separated into three steps: imputation, analysis, and pooling.

Creating imputations. Imputations can be understood as “predicted values plus noise” (Enders, 2022, p. 266). That is, missing values are replaced by plausible values drawn from an imputation model, so that the resulting completed data sets reflect both the predicted value and uncertainty around this prediction (Murray, 2018). A finite number, usually between $m = 20$ and $m = 100$ (Graham et al., 2007) of imputations is saved, leading to m completed data sets. The imputation model can either be specified jointly for all variables (*joint imputation*) or individually (*fully conditional specification*). Both approaches can be extended to multilevel data structures (Enders et al., 2016), but in this thesis, only joint imputation is considered. In this approach, a single multivariate distribution for all variables with missing values is specified, often the multivariate normal distribution for continuous variables (Enders, 2022, p. 263).

Specifying the imputation model. A crucial point in the choice of a valid imputation model is compatibility with the focal analysis model that is used later on. That is, the imputation model must reflect the key structural features of this analysis model (Grund et al., 2018b; Meng, 1994). At the same time, the imputation model is allowed to be richer than the analysis model. Including additional effects or variables that are not part of the analysis model as *auxiliary variables*[5] is considered beneficial for parameter estimation (Collins et al., 2001; Schafer, 2003). These variables may either be causes of missingness, thereby making the missing at random (MAR) assumption more plausible, or they may correlate with variables that contain missing values, which can increase the precision of the imputations and reduce bias in parameter estimates (Collins et al., 2001).

For a random-intercept analysis model, the *multivariate empty model* generally represents a suitable joint imputation model. In this approach, all level-1 and level-2 variables included in the imputation model are treated as outcomes, and their variances and covariances are decomposed into between- and within-group components without specifying predictors (Grund et al., 2016b). In this way, the multilevel structure of the data and the associations among variables at the respective levels are preserved. This also provides the relevant within- and between-group information for a random-intercept model with a contextual effect. For the model introduced in Equation 2.4, an appropriate joint imputation model with one additional level-2 variable can be written as:

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix} \tag{2.6}$$

W_j represents a level-2 variable and therefore has no level-1 residual term. If W_j is not included in the analysis model, it can serve as an auxiliary variable. For example, in the musical self-concept study, the years of teaching experience of the ensemble teachers might be available. Although this variable is not part of the subsequent analysis model, it may correlate with practice behavior or musical self-concept and could therefore serve as an auxiliary level-2 variable. The random effects and residuals are assumed to follow multivariate normal distributions:

$$\begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} \sim N(0, \Psi), \qquad \begin{pmatrix} e_{Yij} \\ e_{Xij} \end{pmatrix} \sim N(0, \Sigma)$$

where the covariance matrix Ψ captures the between-group variability and covariation of the random intercepts and Σ represents the within-group covariance structure of the level-1 residuals:

$$\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}.$$

Analyzing the completed data sets and pooling the results. After the imputation step, researchers are in the comfortable position to have m completed data sets that can be analyzed using standard frequentist methods. To each of the data

[5]Auxiliary variables are variables whose only purpose is to improve the missing-data imputation (Collins et al., 2001).

sets, the substantive analysis model of interest (e.g., the random intercept model) is now fitted. Each analysis proceeds according to the standard estimation procedure described in Section 2.1.5, up to the computation of parameter estimates and their standard errors. Hypothesis testing is not conducted at this stage. The result of this step is a set of m parameter estimates and corresponding standard errors. These m estimates and standard errors are aggregated into a single set of results following Rubin's pooling rules (Rubin, 1987). He defined the point estimates of effects as the arithmetic average of the m estimates:

$$\hat{\theta} = \frac{1}{m}\sum_{k=1}^{m}\hat{\theta}_k$$

$\hat{\theta}$ is the pooled point estimate, and $\hat{\theta}_k$ is the estimate from data set k ($k = 1, ..., m$).

For the pooled standard errors, both the within-imputation variance (which is based on a complete data set) and the between-imputation variance (which adds the uncertainty due to missing values) are combined. The within-imputation variance is the arithmetic average of the imputation's variances (standard errors squared):

$$\bar{V}_w = \frac{1}{m}\sum_{k=1}^{m} SE_k^2$$

The between-imputation variance is defined as the variance of point estimates between imputations that is due to uncertainty caused by the missing values:

$$V_b = \frac{1}{m-1}\sum_{k=1}^{m}(\hat{\theta}_k - \hat{\theta})^2$$

The total pooled standard error combines the within-imputation variance $\bar{V}_w$ and the between-imputation variance V_b:

$$SE = \sqrt{\bar{V}_w + V_b + \frac{V_b}{m}}$$

The third term in this expression acts as a correction factor for using a finite number of imputations (Enders, 2022, p. 284). As the number of imputations m increases, this correction term approaches zero. With an appropriate point estimate and standard error, the remaining step (see Figure 2.2) is the choice of the sampling distribution. Rubin proposed a calculation method for degrees of freedom of the t-distribution, that is based on the relative increase of variance due to missing values RIV ($RIV = \frac{(1+1/m)V_b}{\bar{V}_w}$) and the number of imputations m:

$$df_R = (m-1)\left(1 + \frac{1}{RIV^2}\right)$$

This formula assumes that the degrees of freedom of the complete data are effectively infinite (Barnard & Rubin, 1999). In smaller samples, the classic Rubin formula therefore tends to produce degrees of freedom that are too large, sometimes even larger than the degrees of freedom of the complete data. Barnard and Rubin (1999) thus proposed a small-sample-adjusted degrees-of-freedom calculation for pooled parameters. Their formula downweights the complete-data degrees of freedom to account for the uncertainty due to missing values. As the complete-data degrees of freedom cannot easily be obtained for multilevel analyses, a suitable degrees-of-freedom approximation must be chosen for the analysis. Possible approaches include the methods described in Section 2.2, but a specific recommendation for multilevel models does not exist.

2.3.2.2 Full Information Maximum Likelihood (FIML)

Full information maximum likelihood (FIML) is a model-based approach in which parameter estimates are obtained directly by maximizing the likelihood of the observed data (Enders, 2022, p. 98). Under the assumption that data are missing at random (MAR), FIML yields unbiased and efficient parameter estimates (McNeish, 2017a). Incorporating auxiliary variables is less straightforward in FIML than in multiple imputation. Nevertheless, several approaches have been proposed to include auxiliary information in FIML estimation, such as the saturated correlates model or the extra dependent variable approach (Graham, 2003), as well as the factored regression framework described below for the Bayesian missing-data approaches.
For the random intercept model with group-mean effects considered in this thesis, FIML is not always well suited when missingness occurs in predictor variables. In such cases, maximum likelihood estimation requires additional distributional assumptions about the distribution of the predictors in order to incorporate them into the likelihood function (Grund et al., 2018b). In multilevel models with manifest group means, these assumptions may lead to biased estimates of between-group effects (Grund et al., 2019).

2.3.2.3 Fully Bayesian Estimation

A Bayesian framework can naturally be applied to missing-data problems. In contrast to the two-step procedure of multiple imputation, Bayesian missing-data analysis directly uses the posterior distribution of the model parameters conditional on the observed data for statistical inference. Missing values are treated as additional unknown quantities. To simulate these values during estimation, a model for the variables with missing data must be specified. A convenient way to specify the joint distribution of the outcome and the variables with missing values is to apply the probability chain rule and express the joint distribution as a product of conditional distributions, which is referred to as *sequential specification* or *factored regression specification* (Lüdtke et al., 2020):

$$g\left(y,\ x;\gamma\right) = g_y\left(y\middle|x;\gamma_y\right) \times g_x(x;\gamma_x)$$

where $\gamma = (\gamma_y, \gamma_x)$ denotes the vector of all model parameters. Here, the joint density is decomposed into two components. The first term $g_y(y \mid x; \gamma_y)$ corresponds to the density implied by the analysis model, describing the conditional distribution of the outcome given the covariate X. The second term $g_x(x; \gamma_x)$ specifies a model for the distribution of the covariate X. Each conditional distribution in a sequential modeling framework is typically specified as a regression model. Consequently, the implied joint distribution is constructed from a sequence of conditional models, each imposing its own distributional assumptions.

An advantage of this formulation is its flexibility: it allows the inclusion of variables with different measurement scales and facilitates the use of auxiliary variables that are not part of the substantive analysis model (Daniels et al., 2014; Erler et al., 2016). For example, auxiliary variables such as W_j may be included to improve the imputation of missing values while the substantive model of interest is still estimated without W_j. For this purpose, auxiliary variables can be placed in the first position of the factored regression, so that the analysis variables predict the auxiliary variables (Enders, 2022, p. 215):

$$g\left(y,\ x,w;\gamma\right) = g_w\left(w\middle|y,x;\gamma_w\right) \times g_y\left(y\middle|x;\gamma_y\right) \times g_x(x;\gamma_x) \tag{2.7}$$

Because all missing values are resampled in each iteration, the uncertainty due to missing data is automatically included in the posterior distribution (Erler et al., 2016).

2.3.3 Present study

Both challenges considered here — small sample sizes and missing data — often co-occur in empirical research, but missing-data methods have rarely been investigated in small-sample settings. For multiple imputation, some studies suggest that small-sample adjustments applied during pooling may improve inferential performance in both single-level and multilevel contexts (McNeish, 2017a, 2018). However, these adjustments have not been compared to standard inferential approaches based on unadjusted degrees of freedom, and the models considered did not incorporate contextual predictors. Fully Bayesian estimation may offer a promising model-based alternative to MI, as it naturally accommodates both missing predictor values and small sample sizes. It is often recommended in more complex multilevel models, for example in the presence of nonlinearities or interaction effects (Depaoli & Clifton, 2015; Enders et al., 2020; Muthén & Asparouhov, 2012). However, it may also represent a suitable alternative to likelihood-based approaches such as FIML in simpler models with contextual effects, where the handling of missing predictor variables and the construction of group means may lead to biased estimates (Grund et al., 2019). To evaluate this, the present study uses a simulation design to compare methods for handling missing data in a predictor variable in a small-sample multilevel random intercept model. In particular, it examines whether applying a small-sample-adjusted degrees-of-freedom approximation with the $N_2 - q - 1$ heuristic (Equation 2.5) after multiple imputation improves inferential performance, and whether fully Bayesian estimation provides a suitable alternative to imputation-based or likelihood-based approaches.

Simulation studies can be understood as computer experiments, in which data with known characteristics are generated and then analysed using the methods of interest (Morris et al., 2019). Because the true parameter values are known, the resulting estimates can be compared with the truth, allowing the performance of the methods to be evaluated (Siepe et al., 2024). The simulation design and evaluation criteria are described in the following chapter.

Chapter 3

Methods

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

3.1 Data generation mechanism

3.1.1 Data-generating model

The data-generating model was a two-level random-intercept model with with a contextual effect of X. Specifically, the level-1 predictor X_{ij} was decomposed into person-level deviations from the group mean, $(X_{ij} - \bar{X}_j)$, and the manifest group mean, $\bar{X}_j$, at level 2. In addition, the model included one level-2 predictor W_j:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_j) + \gamma_{01}\bar{X}_j + \gamma_{02}W_j + u_{0j} + e_{ij} \tag{3.1}$$

Here, γ_{10} represents the within-group effect of X, γ_{01} the between-group effect of X and γ_{02} the effect of W on Y. The random effects at the group and individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \qquad e_{ij} \sim N(0, \sigma^2),$$

and were assumed to be independent. The variance components ψ^2 and σ^2 were chosen such that Y_{ij} had a population variance of 1 and a prespecified residual intraclass correlation coefficient, as described in Appendix A. The level-1 predictor X_{ij} was simulated by from independent between- and within-group components:

$$X_{ij} = u_{Xj} + e_{Xij},$$

with

$$u_{X_j} \sim N(0, {\tau_x}^2), \qquad e_{X_{ij}} \sim N(0, {\sigma_x}^2)$$

The variances ${\tau_x}^2$ and σ_x^2 were chosen to achieve a prespecified intraclass coefficient for X (see Section 3.1.3), with ${\tau_x}^2 = ICC_x$ and ${\sigma_x}^2 = 1 - ICC_x$. This parameterization ensured that X_{ij} had a population mean of 0 and variance of 1. The resulting predictor was decomposed into the person-level deviation from the group mean, $(X_{ij} - \bar{X}_j)$, and the manifest group mean, $\bar{X}_j$, both of which entered the data-generating model.

The level-2 predictor W_j was generated using the following linear model:

$$W_j = a\bar{X}_j + e_{Wj} \qquad e_{Wj} \sim N(0, {\sigma_w}^2)$$

Since $\bar{X}_j$ and e_{Wj} both have expectation 0, W_j also has population mean 0. The parameters a and σ_w^2 were chosen to obtain population variance 1 and a prespecified

TABLE 3.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

Data conditions	
n	10
N_2	15 / 30 / 60
ICC of X and Y	.1 / .3
Correlation of $\bar{X}$ and W	.3
Model parameters	
Effect of $(X_{ij} - \bar{X}_j)$	0.2
Effect of $\bar{X}_j$	0 / 0.4
Effect of W_j	0
Missing values	
Pattern	$X \sim W$
Mechanism	MCAR / MAR
Proportion	30%

population correlation between W_j and $\bar{X}_j$, denoted by $\rho_{\bar{X}W}$.[1]

3.1.2 Missing data generation

After simulation of the complete dataset, values of X were deleted to induce missingness. A 'missingness tendency' r_{ij} was simulated for each X_{ij}-value according to

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \qquad e_{ij} \sim N(0, 1 - \beta^2)$$

The resulting r_{ij}-values are normally distributed. Observations with $r_{ij} > 0$ were set to missing. The intercept α controls the overall proportion of missing values in X, whereas the slope β determines the strength of the MAR mechanism. When $\beta = 0$, missingness is MCAR. With increasing β, the missingness mechanism becomes more dependent on the W_j-values; with $\beta = 1$, missingness would be completely deterministic. For the present simulation, $\beta = .3$ was used for MAR conditions, corresponding to approximately 9% explained variance in the latent missingness tendency.

3.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table 3.1.

The number of level-2 units was manipulated across conditions ($N_2 = 15, 30, 60$), while the number of level-1 observations was kept constant ($n = 10$). The between-group effect of $\bar{X}_j$ was varied ($\gamma_{01} = 0, 0.4$), whereas the within-group effect of $(X_{ij} - \bar{X}_j)$ was held constant at 0.2 and the effect of W_j was set to zero. This results in conditions with either a negative or a positive context effect (i.e., $\gamma_{10} - \gamma_{01}$) of X in the population model. The ICC of X and the residual ICC of Y were varied ($\rho_X = \rho_{Y|X} = .1, .3$). Missing data were generated on X with a varying missingness mechanism (MCAR or MAR with a strength of $\beta = .3$), with a constant missingness rate of 30%. The factors were combined fully factorially, creating $3 \times 2 \times 2 \times 2 = 24$ conditions.

Taken together, the simulation design targets conditions that are particularly challenging for multilevel estimation (small level-2 sample sizes and low ICCs), while also

[1]The parameters were set to $\sigma_w^2 = 1 - \rho_{\bar{X}W}^2$ and $a = \rho_{\bar{X}W} / \sqrt{\tau_x^2 + \sigma_x^2/n}$, which yields $Var(W_j) = 1$ and $Cor(W_j, \bar{X}_j) = \rho_{\bar{X}W}$; see Appendix A for the derivation.

covering less challenging scenarios for comparison. The between-group effect of $\bar{X}_j$ is varied to evaluate method performance under zero versus a non-zero effect.

3.2 Estimands of the focal analysis model

The focal analysis model was a random-intercept model with a contextual effect of X, corresponding to the data-generating model but omitting the additional level-2 predictor W_j:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_j) + \gamma_{01}\bar{X}_j + u_{0j} + e_{ij} \tag{3.2}$$

The primary target was the level-2-effect, γ_{01}. Additionally, the level-1-effect γ_{10} was evaluated.

3.3 Methods of Missing Data Handling

Four methods were be compared, alongside the benchmark analysis of complete data (CD).

3.3.1 Listwise deletion (LD)

LD was used as a standard reference method for missing data handling. Cases with missing X_{ij}-values were excluded prior to analysis. The group means $\bar{X}_j$ were then calculated only from the remaining observations. Then, group-mean centering was applied.

3.3.2 Multiple imputation (MI) with standard (MI-R) and adjusted (MI-a) degrees of freedom

For the two multiple imputation methods, a joint modeling approach was employed. The imputation model was the multivariate empty model specified before (Equation 2.6), with W_j as an auxiliary variable on level 2. Imputation was executed with the R package `mitml` (Grund et al., 2015) which provides an interface to the `jomo` package (Quartagno & Carpenter, 2023). The algorithm was run with 2000 burn-in-iterations and 3000 post-burn-in iterations, saving every 300th imputation, resulting in 10 imputations.
For MI-R, degrees of freedom for calculating the test statistic were calculated with the software standard (Rubin´s rules, assuming infinite degrees of freedom in complete data). For MI-a, complete data degrees of freedom were calculated with $df_{Lv1} = N_1 - 3$ and $df_{Lv2} = N_2 - 2$, using the heuristic from Equation 2.5.

3.3.3 Fully Bayesian Analysis

A factored regression specification with the auxiliary variable W_j was used within the Bayesian framework that corresponds to the model described in Equation 2.7. The burn-in period was set to 1000 iterations, followed by 3000 post-burn-in iterations. The model was estimated using the package `mdmb` (Robitzsch & Luedtke, 2024) with its default noninformative priors.

3.3.4 Extracted quantities

For complete data, listwise deletion and the two multiple imputation methods, point estimates of γ_{10} and γ_{01} as well as their standard errors and 95% confidence intervals were extracted. For the fully Bayesian method, the posterior means of γ_{10} and γ_{01} were used as point estimates along with posterior standard deviations and 95% credible intervals. For comparability with the frequentist approaches and in line with common simulation practice, these posterior summaries were treated analogously to frequentist estimates and intervals (Morris et al., 2019).

3.4 Performance measures

Performance was evaluated using standard measures for simulation studies (Morris et al., 2019; Siepe et al., 2024), namely bias, coverage of confidence/credible intervals, empirical standard deviations of the estimates across simulation replications, and average width of confidence/credible intervals. Monte Carlo uncertainty of the estimated performance measures was quantified using Monte Carlo standard errors (MCSEs), based on the formulas provided by Siepe et al. (2024). Intervals of ± 2 MCSE, approximately corresponding to 95% confidence intervals for the estimated performance measures, were used to guide the interpretation of the results. A bias of 10% of the true parameter value was considered substantial for non-zero effects. For a true null effect, bias was considered substantial if it lay outside the interval of 0 ± 2 MCSE. For coverage, estimates whose ± 2 MCSE interval included the nominal level of 95% were interpreted as compatible with nominal coverage. Estimates above this range were interpreted as indicating overcoverage, whereas estimates below this range were interpreted as indicating undercoverage. Empirical standard deviations and interval widths were interpreted comparatively, as neither measure has a fixed nominal target value. Smaller empirical standard deviations indicate higher precision of point estimates, and narrower confidence/credible intervals indicate higher inferential precision. However, these patterns were considered favorable only when bias was negligible and coverage was close to the nominal level.

Monte Carlo uncertainty of the estimated performance measures was quantified using Monte Carlo standard errors (MCSE), using the formulas provided in Siepe et al. (2024). Intervals of ± 2 MCSE, approximately corresponding to 95% confidence intervals for the estimated performance measures, were used to guide the interpretation of the results. For each condition, 1000 simulation replications were conducted, in line with comparable simulation studies (Enders et al., 2016; Grund et al., 2018b) and resulting in sufficiently small MCSEs for coverage. Convergence of the MCMC algorithm was evaluated under a challenging condition, characterized by a low ICC and a small level-2 sample size, by inspecting $\hat{R}$ values (Gelman & Rubin, 1992), trace plots, and autocorrelation plots. These diagnostics indicated satisfactory convergence, with no problematic autocorrelation; see Appendix B for details. All simulations were conducted in R (version 4.5.2, R Core Team, 2025) using a reproducible workflow (e.g., fixed random seeds and controlled package environments). The full code and all relevant materials to reproduce the results are publicly available via Zenodo (https://doi.org/10.5281/zenodo.19545308)[2].

[2]this is only a test release right now

Chapter 4

Results

For multiple imputation, bias and empirical standard deviations are identical for the MI-R and MI-a variants, because both approaches are based on the same pooled point estimates and differ only in the degrees-of-freedom calculation used for inference. Therefore, MI-R and MI-a are reported separately only for coverage and confidence interval width.

For the level-1 effect γ_{10}, bias was consistently negligible relative to the corresponding MCSEs. Coverage, empirical standard deviations, and confidence/credible interval widths were comparable across methods and closely aligned with the corresponding CD results. Results are therefore reported here only for the level-2 effect γ_{01}. Full results for all parameters of interest and performance measures are provided in Appendix C.

4.1 Bias of point estimation

The findings for γ_{01} are depicted in Figure 4.1 for all conditions with missingness at random (MAR). Results under MCAR followed very similar patterns and are therefore not shown separately.

LD resulted in substantial bias, especially in conditions with low ICC. For a true null effect, estimates were positively biased, with a maximum of 0.042. For a positive level-2 effect, estimates were negatively biased, with a maximum of -0.050. The magnitude of bias was similar across sample sizes and did not differ markedly between MAR and MCAR conditions.

MI resulted in larger bias than LD in several low-ICC conditions with a substantial true effect, where bias ranged from -0.044 to -0.078, corresponding to approximately 10-20% of the true effect[1]. In all other conditions, bias estimates for MI were small relative to their MCSEs (all $|\text{bias}| \leq 0.02$).

In contrast, the Bayesian approach showed only small bias across all simulation conditions except the most challenging conditions ($N_2 = 15$, $ICC = .1$, $\gamma_{01} = 0.4$), where bias was -0.039 (MAR) and -0.036 (MCAR). These values remained below 10% of the true effect. All other biases were small ($|\text{bias}| \leq 0.015$).

4.2 Coverage of confidence and credible intervals

The results for coverage of the confidence and credible intervals for γ_{01} are shown for $N_2 = 15$ in Figure 4.2, as larger sample sizes exhibited similar but attenuated patterns. Overall, coverage rates were close to the nominal level (95%) for $N_2 = 30$ and $N_2 = 60$ across all methods. LD yielded coverage rates close to the nominal level in

[1]However, in a sensitivity analysis, the use of alternative weakly informative priors in these conditions consistently reduced bias by approximately 30–60% (see Appendix D for details).

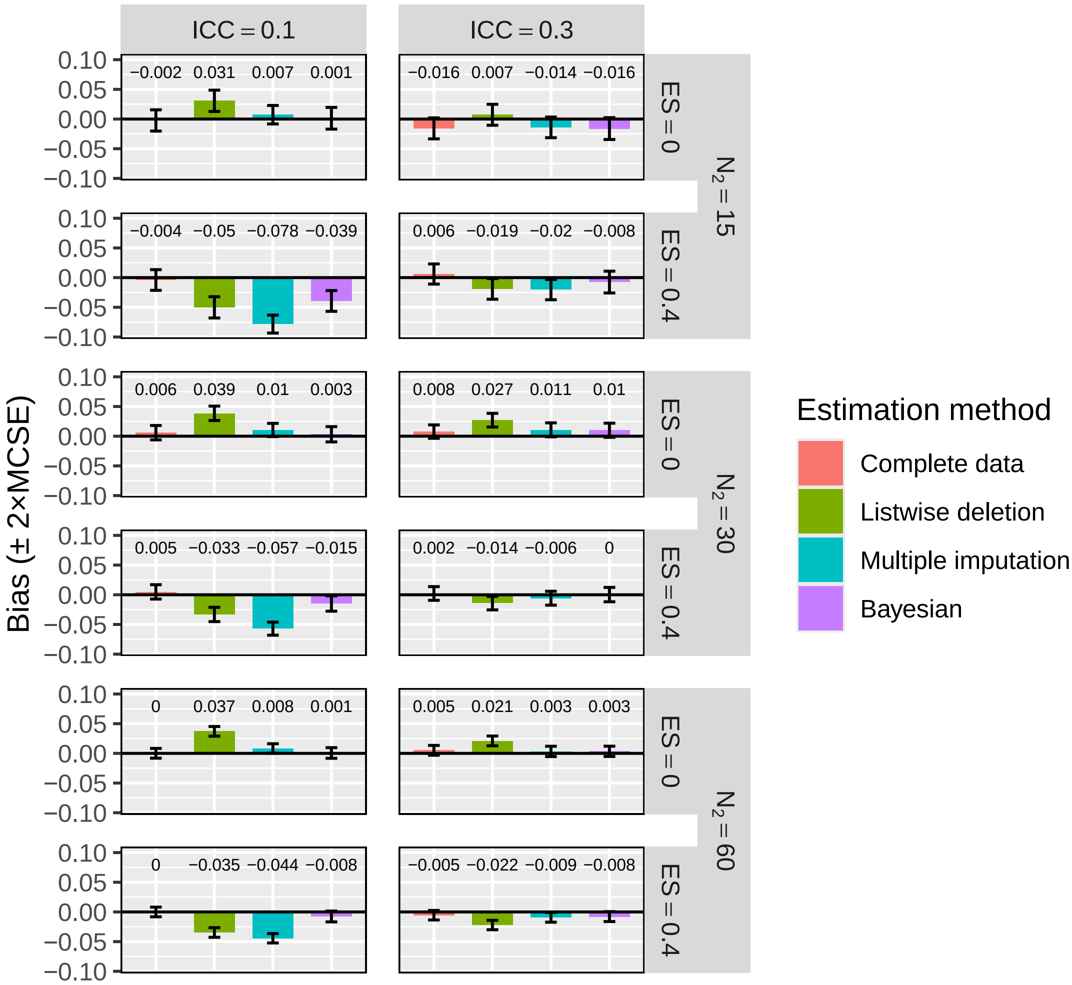

FIGURE 4.1: Bias of the between-group effect γ_{01} across simulation conditions under missing at random (MAR). Error bars represent ±2 Monte Carlo standard errors (MCSE). ES = effect size; ICC = intraclass correlation coefficient used for both the residual ICC of Y and the ICC of X; N_2 = number of groups.

all conditions. For multiple imputation, coverage in the MI-R variant was close to the nominal level in high-ICC conditions, with a tendency toward overcoverage in low-ICC conditions. For $ICC = .1$, coverage ranged from 0.952 ($N_2 = 30$, $\gamma_{01} = 0.4$, MAR) to 0.971 ($N_2 = 60$, $\gamma_{01} = 0$, MCAR). The approach using small-sample adjusted degrees of freedom resulted in increased overcoverage compared to the standard degrees of freedom, particularly for $N_2 = 15$. Overcoverage was again more pronounced in low-ICC conditions, ranging from 0.960 ($N_2 = 60$, $\gamma_{01} = 0.4$, MCAR) to 0.986 ($N_2 = 15$, $\gamma_{01} = 0$, MCAR) for $ICC = .1$.
The Bayesian approach exhibited overcoverage for $N_2 = 15$, ranging from 0.963 ($\gamma_{01} = 0$, MCAR, $ICC = .1$) to 0.985 ($\gamma_{01} = 0.4$, MCAR, $ICC = .1$). This method yielded the highest (most conservative) coverage rates at the smallest level-2 sample size, except in the condition with MCAR, $ICC = .1$, and $\gamma_{01} = 0$. With increasing level-2 sample size, this pattern attenuated. For $N_2 = 30$, coverage estimates were largely within the bounds of Monte Carlo error, and for $N_2 = 60$, they fell entirely within these bounds across all conditions.

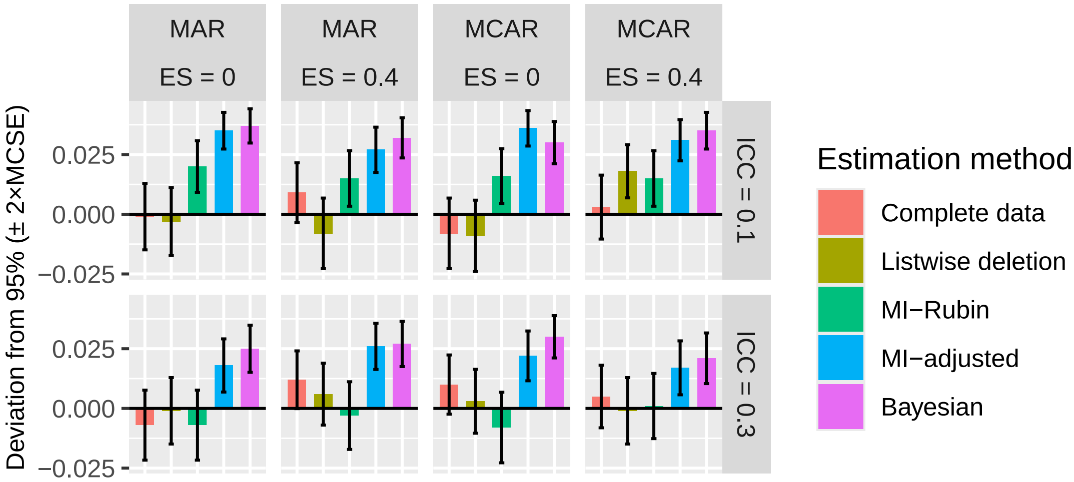

FIGURE 4.2: Deviation from nominal (95%) coverage of confidence/credible intervals across methods and simulation conditions for $N_2 = 15$. Error bars represent ±2 Monte Carlo standard errors (MCSE). ES = effect size; MAR = missing at random; MCAR = missing completely at random; ICC = intraclass correlation coefficient used for both the residual ICC of Y and the ICC of X; MI-Rubin = multiple imputation using Rubin's degrees of freedom; MI-adjusted = multiple imputation using adjusted degrees of freedom.

4.3 Standard deviation of point estimates

Standard deviations of point estimates across conditions and methods are illustrated in Figure 4.3 for MAR, as differences between missingness mechanisms were small.

LD produced standard deviations very close to those obtained under CD, ranging from 0.124 ($N_2 = 60$, MCAR, $\gamma_{01} = 0.4$) to 0.300 ($N_2 = 15$, MCAR, $\gamma_{01} = 0$). For MI, standard deviations were comparable to CD across conditions, with a tendency toward smaller values in some low-ICC conditions. For example, for $N_2 = 15$, $\gamma_{01} = 0$, and MAR missingness, MI yielded a standard deviation of 0.247, compared to 0.283 under CD. The Bayesian approach yielded standard deviations that were generally slightly larger than those from complete-data analysis, typically by approximately 0.006 to 0.020. Standard deviations were generally smaller under high ICC conditions than under low ICC conditions in the Bayesian framework, except in the conditions with $\gamma_{01} = 0.4$ and $N_2 = 15$. In contrast, MI consistently yielded slightly larger standard deviations under high-ICC conditions. Although these differences were small, they were larger than would be expected from Monte Carlo error alone.

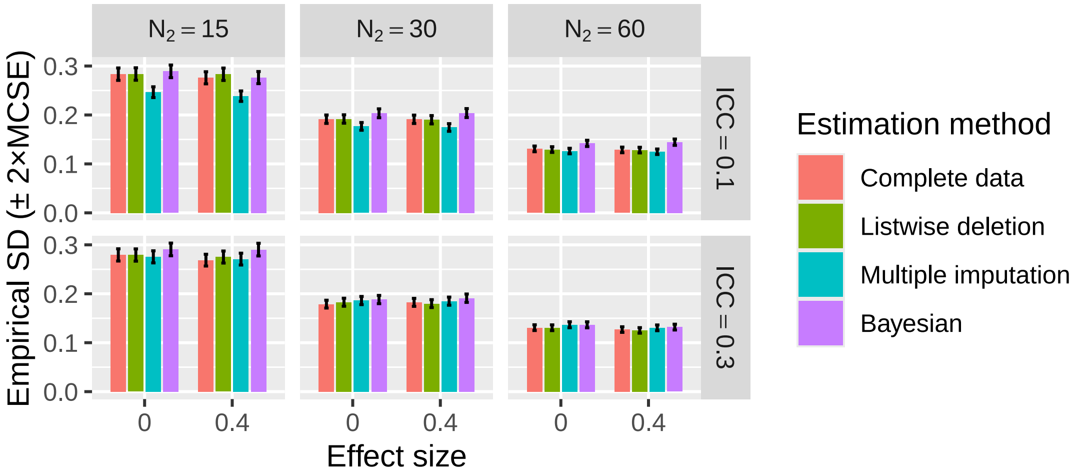

FIGURE 4.3: Empirical standard deviation (SD) of point estimates for γ_{01} for conditions with missingness at random (MAR). Error bars represent ±2 Monte Carlo standard errors (MCSE). N_2 = number of groups; ICC = intraclass correlation coefficient used for both the residual ICC of Y and the ICC of X.

4.4 Width of confidence and credible intervals

Results for the width of confidence and credible intervals are illustrated in Figure 4.4, as similar patterns were observed across conditions. Overall, the Bayesian approach yielded the widest credible intervals, followed by multiple imputation with adjusted degrees of freedom (MI-a). For $N_2 = 15$, the MI-R variant produced interval widths that were slightly smaller than those obtained under CD (e.g., 1.115 vs. 1.170 for MCAR, $ICC = .1$, $\gamma_{01} = 0$). For $N_2 = 30$ and $N_2 = 60$, MI-R interval widths were slightly larger than those from CD, while still remaining smaller than those from MI-a and the Bayesian approach. Confidence intervals from LD were similar to those obtained with CD across conditions.
At the smallest level-2 sample size, $N_2 = 15$, differences between methods were most pronounced. For example, when $\gamma_{01} = 0$ and $ICC = .1$ under MCAR, interval width for γ_{01} was 1.115 under MI-R, 1.284 with MI-a and 1.456 under the Bayesian approach. With increasing level-2 sample size, differences between methods became less pronounced, although interval widths for the Bayesian approach remained slightly larger. For instance, when $\gamma_{01} = 0$, $ICC = .1$, and MCAR at $N_2 = 60$, interval width was 0.517 for complete-data analysis, 0.520 for LD, 0.545 for MI-R, 0.559 for MI-a, and 0.582 for the Bayesian approach. Within the fully Bayesian approach, interval widths were generally slightly smaller under high-ICC conditions than under low-ICC conditions. In contrast, the other methods did not show a consistent ICC-related pattern.

4.5 Summary of results

The methods showed different performance profiles for the level-2 effect. LD resulted in increased bias, while other performance measures were generally similar to CD results. MI produced substantial downward bias in the estimation of the level-2 effect

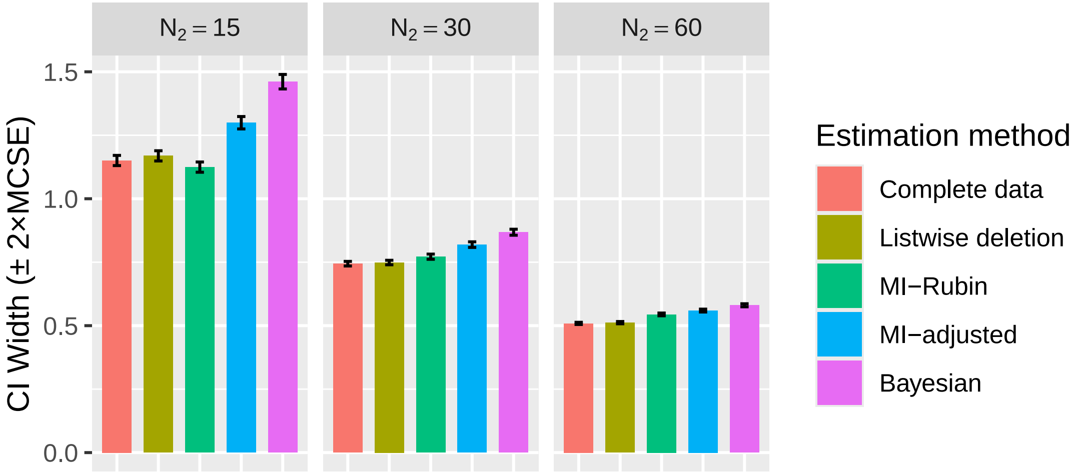

FIGURE 4.4: Width of confidence/credible intervals for $\gamma_{01} = 0.40$ under missing at random (MAR) and $ICC = .10$, where ICC denotes the intraclass correlation coefficient used for both the residual ICC of Y and the ICC of X. Error bars represent ± 2 Monte Carlo standard errors (MCSE). N_2 = number of groups; MI-Rubin = multiple imputation with Rubin's degrees of freedom; MI-adjusted = multiple imputation with small-sample-adjusted degrees of freedom.

under conditions of low ICC and MAR missingness. At the same time, the standard variant (MI-R) yielded coverage rates close to the nominal level across conditions. Applying a small-sample correction to the degrees of freedom led to increased coverage beyond the nominal level and wider confidence intervals. The fully Bayesian approach showed small to negligible bias across conditions, but resulted in the widest intervals among all methods and a tendency toward overcoverage in the smallest sample size.

Chapter 5

Discussion

This thesis investigated the performance of two modern methods for handling missing predictor data in small-sample multilevel models, MI and a fully Bayesian approach, alongside LD as a commonly used default. It focused on a reduced setting with a single continuous predictor which was included at level 1 as an individual-level predictor and at level 2 as a manifest group-mean predictor, within a random intercept model. In addition, one auxiliary variable at level 2 was included to explain missingness under a MAR mechanism, without being directly related to the outcome.

The results indicate a trade-off between bias and interval width when comparing MI and the fully Bayesian approach. In low-ICC settings with default prior specifications, the fully Bayesian approach showed less bias than MI, and this difference persisted even at the largest number of groups considered in the present design. Although MI can be understood as an approximate Bayesian procedure, as it relies on draws from the posterior predictive distribution (e.g., Enders, 2022, p. 299; Schafer & Graham, 2002), the present results suggest that this connection does not necessarily translate into comparable performance in small samples. In particular, the sensitivity analysis indicates that prior specification may play an important role. This analysis focused on the conditions in which MI showed the strongest bias. In these conditions, replacing the default prior specification for the level-2 covariance matrix with a weakly informative prior based on plausible values substantially reduced bias in the level-2 effect estimate, while coverage remained close to the nominal level. This suggests that when reasonable prior information is available for level-2 variance components, incorporating this information through weakly informative priors may improve the performance of MI in comparable settings with few groups without necessarily compromising coverage.

Another aspect of MI that was specifically investigated in this thesis is the behavior of inference based on uncorrected versus small-sample-adjusted degrees of freedom. The findings suggest that, in this setting, standard Rubin's rules already yield approximately valid inference, and that additional small-sample corrections may not be necessary. This is somewhat surprising, as it is commonly recommended to use an appropriate sampling distribution in small samples to avoid overly narrow confidence intervals; a recommendation which also applies to multilevel analyses (Kenward & Roger, 1997; Schaalje et al., 2002) and within MI (Barnard & Rubin, 1999). However, confidence intervals are influenced by multiple sources of uncertainty beyond the degrees-of-freedom approximation alone. In particular, REML estimation was used for analyzing the imputed data sets. Compared to ML, REML typically yields larger estimates of level-2 variance components, which have been shown to improve variance estimation in small samples (Elff et al., 2021; McNeish & Stapleton, 2016b). As these variance components directly affect the standard errors of the fixed effects, they also influence the within-imputation variance and, consequently, the total variance used for inference. This may have contributed to wider intervals and thus higher coverage.

In addition, MI was performed using joint modeling via `jomo`. Previous work has shown that inverse Wishart priors with few degrees of freedom, as implemented in `jomo`, can introduce additional variability into fixed-effect estimates (Audigier et al., 2018). Overall, this finding may be reassuring for applied researchers relying on default degrees-of-freedom specifications.
In terms of implementation, both fully Bayesian estimation and MI allow the inclusion of auxiliary variables, which were also part of the present setup. In MI, their inclusion is typically well-documented and supported by standard workflows. In fully Bayesian approaches, incorporating auxiliary variables without altering the analysis model requires additional modeling decisions that may not be as straightforward to practitioners. Against this background, the choice between approaches is likely to depend not only on statistical performance but also on the research context and analytical goals. MI may be advantageous when several analyses, multiple outcomes, or model comparisons are planned. Because key decisions about auxiliary variables and the operationalization of the MAR assumption are made once at the imputation stage and then carried forward across subsequent analyses, MI may provide a comparatively simple and less error-prone workflow than model-based approaches in which the missing data treatment must be specified separately for each analysis. This is relevant because decisions about auxiliary variables are not merely technical; their inclusion can affect both bias and efficiency of estimation (Yuan & Savalei, 2014). In contrast, fully Bayesian approaches may be particularly useful in cumulative research settings, where prior information can be updated across studies and evidence can be built up and integrated over time (Van De Schoot et al., 2014). However, these distinctions are not absolute, and both approaches can, in principle, be adapted to a wide range of analytical scenarios.

This simulation study comes with several limitations. Firstly, more complex model structures may behave differently under the investigated methods. The present study focused on a relatively simple model without nonlinear terms, interaction effects, or random slopes. Sample size requirements in multilevel models generally depend on model complexity, as more complex specifications involve a larger number of parameters that must be estimated reliably (Hox & McNeish, 2020). Therefore, what worked well in the present study might be more problematic in more complex models. In particular, when such features are present, the treatment of missing data becomes substantially more challenging (Enders et al., 2016; Grund et al., 2016a, 2018b). For MI, default imputation models often lead to incompatibility between the imputation and analysis model and, consequently, biased estimates (Bartlett et al., 2015; Kim et al., 2018; Seaman et al., 2012). Therefore, approaches that allow for substantive-model-compatible MI, such as sequential modeling, have been proposed to address these challenges (Bartlett et al., 2015; Enders et al., 2020; Grund et al., 2021; Lüdtke et al., 2020; Zhang & Wang, 2017), but these approaches have not yet been thoroughly investigated in data sets with few groups. Furthermore, the results only apply to continuous variables. Categorical outcomes often require larger samples for stable estimation, and common corrections for linear mixed models do not directly extend to generalized mixed models (Hox & McNeish, 2020; McNeish, 2016a; Moineddin et al., 2007). In addition, MI is more complex when incomplete variables are categorical rather than continuous, because binary, nominal, and ordinal variables require different imputation models and problems such as perfect prediction may occur (Van Buuren, 2018; White et al., 2011).

Second, the present results may not generalize to more extreme small-sample settings. With very few groups, for example fewer than 10, variance components are difficult to estimate reliably, which can affect both MI and Bayesian approaches (Hox

& Maas, 2001; Maas & Hox, 2005; McNeish & Stapleton, 2016a). Furthermore, group size was fixed at 10 for each group. The present study therefore cannot determine whether different patterns of results would be obtained with very small or highly unbalanced group sizes (i.e., unequal number of observations per group). Very small group sizes have been associated with upward bias in level-2 variance components when the number of groups is also small (Clarke, 2008; McNeish, 2014). Conversely, larger groups have been suggested to partly mitigate the consequences of a small number of groups for some aspects of estimation (Hox & Maas, 2001; Maas & Hox, 2005), an issue that could not be examined in the present design. Evidence on the effects of unbalanced group sizes is sparse, but existing work suggests that imbalance can affect statistical inference, for example by leading to underestimated standard errors of fixed effects, particularly when combined with heteroscedasticity and a small number of groups (Korendijk et al., 2008; Raudenbush & Bryk, 2010, p. 281). Unbalanced group sizes may also have implications for the imputation design in contextual models. In such settings, imputation models that rely on manifest group means and do not account for group size may introduce additional bias (Grund et al., 2018a).

Third, prior specification was only examined in a limited sensitivity analysis for MI in the present study. In complete-data settings, previous research suggests that weakly informative priors often outperform noninformative priors in Bayesian multilevel models with few groups (Gelman, 2006; McNeish, 2016b; Smid et al., 2020), whereas findings on data-dependent priors remain inconclusive (Depaoli, 2013; McNeish, 2016c; Van Erp et al., 2018). However, studies on the role of prior specification in the presence of missing data and few groups remain sparse. For MI in particular, standard software priors are often used, although some findings suggest that prior choice can substantially influence results in MI as well (Audigier et al., 2018; Grund et al., 2018b). Extending this line of research to settings with missing data would therefore be an important direction for future research.

Fourth, several key modeling choices were not systematically varied in the present study. Future research could manipulate estimation methods (ML versus REML), imputation frameworks (joint modeling versus fully conditional specification), small-sample corrections and degrees-of-freedom approximations, prior specifications and model complexity to disentangle their relative contributions to estimation performance. Such work would help clarify the behavior of MI in comparison with fully Bayesian estimation and identify the conditions under which small-sample corrections for degrees of freedom are useful, as well as those under which they may not be necessary. For instance, existing evidence suggests that degree-of-freedom corrections might be beneficial in more complex models, such as growth models with random slopes and longitudinal missingness patterns (McNeish, 2018).

Fifth, two distinct software packages with different default prior specifications were used for MI and Bayesian estimation. Using a common software framework for both approaches, as for example possible with `mdmb` (Grund et al., 2021; Robitzsch & Luedtke, 2024), could improve comparability and help isolate differences attributable to modeling choices rather than implementation details.

Finally, a further limitation concerns the missing data mechanism. The present study assumed missingness under MCAR and MAR mechanisms, whereas the combination of few groups and missing data should also be investigated under MNAR mechanisms, which require different modeling approaches (Du et al., 2022; Enders, 2025). Moreover, the present missingness pattern with missing values only in one predictor variable is very simplified, as in real-world applications, missing values typically occur across multiple variables, including outcomes, level-1 and level-2 predictors, and the grouping variable (Van Buuren, 2018). Even if the joint modeling approach used in this thesis can accommodate multivariate missingness, bias and efficiency of the estimates might be differently affected by different model specifications.

Overall, the present findings provide insight into the behavior of missing data methods in small-sample multilevel settings with missing values in a predictor variable, but further research is required to assess their robustness under more complex and realistic conditions.

References

Aarts, E., Verhage, M., Veenvliet, J. V., Dolan, C. V., & Van Der Sluis, S. (2014). A solution to dependency: Using multilevel analysis to accommodate nested data. *Nature Neuroscience*, *17*(4), 491–496. https://doi.org/10.1038/nn.3648

Aguinis, H., Gottfredson, R. K., & Culpepper, S. A. (2013). Best-practice recommendations for estimating cross-level interaction effects using multilevel modeling. *Journal of Management*, *39*(6), 1490–1528. https://doi.org/10.1177/0149206313478188

Anderman, E. M. (2002). School effects on psychological outcomes during adolescence. *Journal of Educational Psychology*, *94*(4), 795–809. https://doi.org/10.1037/0022-0663.94.4.795

Asparouhov, T., & Muthen, B. (2010). *Bayesian analysis using Mplus: Technical implementation* [Technical Report]. https://statmodel.com/download/Bayes3.pdf

Audigier, V., White, I. R., Jolani, S., Debray, T. P. A., Quartagno, M., Carpenter, J., Van Buuren, S., & Resche-Rigon, M. (2018). Multiple imputation for multilevel data with continuous and binary variables. *Statistical Science*, *33*(2). https://doi.org/10.1214/18-STS646

Barnard, J., & Rubin, D. (1999). Small-sample degrees of freedom with multiple imputation. *Biometrika*, *86*(4), 948–955.

Bartlett, J. W., Seaman, S. R., White, I. R., & Carpenter, J. R. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, *24*(4), 462–487. https://doi.org/10.1177/0962280214521348

Clarke, P. (2008). When can group level clustering be ignored? Multilevel models versus single-level models with sparse data. *Journal of Epidemiology and Community Health*, *62*(8), 752–758. https://doi.org/10.1136/jech.2007.060798

Collins, L. M., Schafer, J. L., & Kam, C.-M. (2001). A comparison of inclusive and restrictive strategies in modern missing data procedures. *Psychological Methods*, *6*(4), 330–351. https://doi.org/10.1037/1082-989X.6.4.330

Croon, M. A., & Van Veldhoven, M. J. P. M. (2007). Predicting group-level outcome variables from variables measured at the individual level: A latent variable multilevel model. *Psychological Methods*, *12*(1), 45–57. https://doi.org/10.1037/1082-989X.12.1.45

Daniels, M. J., Wang, C., & Marcus, B. H. (2014). Fully Bayesian inference under ignorable missingness in the presence of auxiliary covariates. *Biometrics*, *70*(1), 62–72. https://doi.org/10.1111/biom.12121

Depaoli, S. (2013). Mixture class recovery in GMM under varying degrees of class separation: Frequentist versus Bayesian estimation. *Psychological Methods*, *18*(2), 186–219. https://doi.org/10.1037/a0031609

Depaoli, S., & Clifton, J. P. (2015). A Bayesian approach to multilevel structural equation modeling with continuous and dichotomous outcomes. *Structural Equation Modeling: A Multidisciplinary Journal*, *22*(3), 327–351. https://doi.org/10.1080/10705511.2014.937849

Du, H., Enders, C., Keller, B. T., Bradbury, T. N., & Karney, B. R. (2022). A Bayesian latent variable selection model for nonignorable missingness. *Multivariate*

Behavioral Research, *57*(2-3), 478–512. https://doi.org/10.1080/00273171.2021.1874259

Elff, M., Heisig, J. P., Schaeffer, M., & Shikano, S. (2021). Multilevel analysis with few clusters: Improving likelihood-based methods to provide unbiased estimates and accurate inference. *British Journal of Political Science*, *51*(1), 412–426. https://doi.org/10.1017/S0007123419000097

Enders, C. K. (2022). *Applied missing data analysis* (2nd ed). Guilford Press.

Enders, C. K. (2025). Missing data: An update on the state of the art. *Psychological Methods*, *30*(2), 322–339. https://doi.org/10.1037/met0000563

Enders, C. K., Du, H., & Keller, B. T. (2020). A model-based imputation procedure for multilevel regression models with random coefficients, interaction effects, and nonlinear terms. *Psychological Methods*, *25*(1), 88–112. https://doi.org/10.1037/met0000228

Enders, C. K., Mistler, S. A., & Keller, B. T. (2016). Multilevel multiple imputation: A review and evaluation of joint modeling and chained equations imputation. *Psychological Methods*, *21*(2), 222–240. https://doi.org/10.1037/met0000063

Erler, N. S., Rizopoulos, D., Rosmalen, J. V., Jaddoe, V. W. V., Franco, O. H., & Lesaffre, E. M. E. H. (2016). Dealing with missing covariates in epidemiologic studies: A comparison between multiple imputation and a full Bayesian approach. *Statistics in Medicine*, *35*(17), 2955–2974. https://doi.org/10.1002/sim.6944

Fahrmeir, L., Kneib, T., Lang, S., & Marx, B. D. (2021). *Regression: Models, methods and applications*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-63882-8

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, *1*(3). https://doi.org/10.1214/06-BA117A

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2025). *Bayesian data analysis third edition (electronic version with errors fixed as of 20 february 2025)* (3rd ed.). Chapman & Hall/CRC.

Gelman, A., & Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science*, *7*(4). https://doi.org/10.1214/ss/1177011136

Graham, J. W. (2003). Adding missing-data-relevant variables to FIML-based structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *10*(1), 80–100. https://doi.org/10.1207/S15328007SEM1001_4

Graham, J. W. (2009). Missing data analysis: Making it work in the real world. *Annual Review of Psychology*, *60*(1), 549–576. https://doi.org/10.1146/annurev.psych.58.110405.085530

Graham, J. W., Olchowski, A. E., & Gilreath, T. D. (2007). How many imputations are really needed? Some practical clarifications of multiple imputation theory. *Prevention Science*, *8*(3), 206–213. https://doi.org/10.1007/s11121-007-0070-9

Grund, S., Lüdtke, O., & Robitzsch, A. (2016a). Multiple imputation of missing covariate values in multilevel models with random slopes: A cautionary note. *Behavior Research Methods*, *48*(2), 640–649. https://doi.org/10.3758/s13428-015-0590-3

Grund, S., Lüdtke, O., & Robitzsch, A. (2016b). Multiple imputation of multilevel missing data: An introduction to the R package pan. *Sage Open*, *6*(4), 2158244016668220. https://doi.org/10.1177/2158244016668220

Grund, S., Lüdtke, O., & Robitzsch, A. (2018a). Multiple imputation of missing data at level 2: A comparison of fully conditional and joint modeling in multilevel designs. *Journal of Educational and Behavioral Statistics*, *43*(3), 316–353. https://doi.org/10.3102/1076998617738087

Grund, S., Lüdtke, O., & Robitzsch, A. (2018b). Multiple imputation of missing data for multilevel models: Simulations and recommendations. *Organizational Research Methods*, *21*(1), 111–149. https://doi.org/10.1177/1094428117703686

Grund, S., Lüdtke, O., & Robitzsch, A. (2019). Missing data in multilevel research. In S. E. Humphrey & J. M. LeBreton (Eds.), *The handbook of multilevel theory, measurement, and analysis.* (pp. 365–386). American Psychological Association. https://doi.org/10.1037/0000115-017

Grund, S., Lüdtke, O., & Robitzsch, A. (2021). Multiple imputation of missing data in multilevel models with the R package mdmb: A flexible sequential modeling approach. *Behavior Research Methods*, *53*(6), 2631–2649. https://doi.org/10.3758/s13428-020-01530-0

Grund, S., Robitzsch, A., & Luedtke, O. (2015). *Mitml: Tools for multiple imputation in multilevel modeling*. Comprehensive R Archive Network. https://doi.org/10.32614/CRAN.package.mitml

Heisig, J. P., & Schaeffer, M. (2019). Why you should *Always* include a random slope for the lower-level variable involved in a cross-level interaction. *European Sociological Review*, *35*(2), 258–279. https://doi.org/10.1093/esr/jcy053

Holtmann, J., Koch, T., Lochner, K., & Eid, M. (2016). A comparison of ML, WLSMV, and Bayesian methods for multilevel structural equation models in small samples: A simulation study. *Multivariate Behavioral Research*, *51*(5), 661–680. https://doi.org/10.1080/00273171.2016.1208074

Hox, J. J., & Maas, C. J. M. (2001). The accuracy of multilevel structural equation modeling with pseudobalanced groups and small samples. *Structural Equation Modeling: A Multidisciplinary Journal*, *8*(2), 157–174. https://doi.org/10.1207/S15328007SEM0802_1

Hox, J. J., & McNeish, D. (2020). Small Samples in Multilevel Modeling. In *Small Sample Size Solutions* (1st ed., pp. 215–225). Routledge. https://doi.org/10.4324/9780429273872-18

Johnson, A. A., Ott, M. Q., & Dogucu, M. (2022). *Bayes rules!: An introduction to applied Bayesian modeling* (1st ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9780429288340

Kackar, R. N., & Harville, D. A. (1981). Unbiasedness of two-stage estimation and prediction procedures for mixed linear models. *Communications in Statistics - Theory and Methods*, *10*(13), 1249–1261. https://doi.org/10.1080/03610928108828108

Kackar, R. N., & Harville, D. A. (1984). Approximations for standard errors of estimators of fixed and random effect in mixed linear models. *Journal of the American Statistical Association*, *79*(388), 853. https://doi.org/10.2307/2288715

Kadane, J. B. (2015). Bayesian methods for prevention research. *Prevention Science*, *16*(7), 1017–1025. https://doi.org/10.1007/s11121-014-0531-x

Kahan, B. C., Forbes, G., Ali, Y., Jairath, V., Bremner, S., Harhay, M. O., Hooper, R., Wright, N., Eldridge, S. M., & Leyrat, C. (2016). Increased risk of type I errors in cluster randomised trials with small or medium numbers of clusters: A review, reanalysis, and simulation study. *Trials*, *17*(1), 438. https://doi.org/10.1186/s13063-016-1571-2

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, *53*(3), 983. https://doi.org/10.2307/2533558

Kim, S., Belin, T. R., & Sugar, C. A. (2018). Multiple imputation with non-additively related variables: Joint-modeling and approximations. *Statistical Methods in Medical Research*, *27*(6), 1683–1694. https://doi.org/10.1177/0962280216667763

Korendijk, E. J. H., Maas, C. J. M., Moerbeek, M., & Van Der Heijden, P. G. M. (2008). The influence of misspecification of the heteroscedasticity on multilevel regression parameter and standard error estimates. *Methodology*, *4*(2), 67–72. https://doi.org/10.1027/1614-2241.4.2.67

Leonhart, R. (2013). *Lehrbuch Statistik: Einstieg und Vertiefung* (3., überarbeitete und erweiterte Auflage). Verlag Hans Huber.

Little, R. J. A., & Rubin, D. B. (2020). *Statistical analysis with missing data* (3rd edition). Wiley. https://doi.org/10.1002/9781119482260

Lüdtke, O., Marsh, H. W., Robitzsch, A., Trautwein, U., Asparouhov, T., & Muthén, B. (2008). The multilevel latent covariate model: A new, more reliable approach to group-level effects in contextual studies. *Psychological Methods*, *13*(3), 203–229. https://doi.org/10.1037/a0012869

Lüdtke, O., Robitzsch, A., & Grund, S. (2017). Multiple imputation of missing data in multilevel designs: A comparison of different strategies. *Psychological Methods*, *22*(1), 141–165. https://doi.org/10.1037/met0000096

Lüdtke, O., Robitzsch, A., Trautwein, U., & Köller, O. (2007). Umgang mit fehlenden Werten in der psychologischen Forschung. *Psychologische Rundschau*, *58*(2), 103–117. https://doi.org/10.1026/0033-3042.58.2.103

Lüdtke, O., Robitzsch, A., & West, S. G. (2020). Regression models involving nonlinear effects with missing data: A sequential modeling approach using Bayesian estimation. *Psychological Methods*, *25*(2), 157–181. https://doi.org/10.1037/met0000233

Maas, C. J. M., & Hox, J. J. (2005). Sufficient sample sizes for multilevel modeling. *Methodology*, *1*(3), 86–92. https://doi.org/10.1027/1614-2241.1.3.86

McNeish, D. (2014). Modeling sparsely clustered data: Design-based, model-based, and single-level methods. *Psychological Methods*, *19*(4), 552–563. https://doi.org/10.1037/met0000024

McNeish, D. (2016a). Estimation methods for mixed logistic models with few clusters. *Multivariate Behavioral Research*, 1–15. https://doi.org/10.1080/00273171.2016.1236237

McNeish, D. (2016b). On using Bayesian methods to address small sample problems. *Structural Equation Modeling: A Multidisciplinary Journal*, *23*(5), 750–773. https://doi.org/10.1080/10705511.2016.1186549

McNeish, D. (2016c). Using data-dependent priors to mitigate small sample bias in latent growth models: A discussion and illustration using Mplus. *Journal of Educational and Behavioral Statistics*, *41*(1), 27–56. https://doi.org/10.3102/1076998615621299

McNeish, D. (2017a). Missing data methods for arbitrary missingness with small samples. *Journal of Applied Statistics*, *44*(1), 24–39. https://doi.org/10.1080/02664763.2016.1158246

McNeish, D. (2017b). Small sample methods for multilevel modeling: A colloquial elucidation of REML and the Kenward-Roger correction. *Multivariate Behavioral Research*, *52*(5), 661–670. https://doi.org/10.1080/00273171.2017.1344538

McNeish, D. (2018). Brief research report: Growth models with small samples and missing data. *The Journal of Experimental Education*, *86*(4), 690–701. https://doi.org/10.1080/00220973.2017.1369384

McNeish, D., & Stapleton, L. M. (2016a). Modeling clustered data with very few clusters. *Multivariate Behavioral Research*, *51*(4), 495–518. https://doi.org/10.1080/00273171.2016.1167008

McNeish, D., & Stapleton, L. M. (2016b). The effect of small sample size on two-level model estimates: A review and illustration. *Educational Psychology Review*, *28*(2), 295–314. https://doi.org/10.1007/s10648-014-9287-x

Meng, X.-L. (1994). Multiple-imputation inferences with uncongenial sources of input. *Statistical Science*, *9*(4). https://doi.org/10.1214/ss/1177010269

Moineddin, R., Matheson, F. I., & Glazier, R. H. (2007). A simulation study of sample size for multilevel logistic regression models. *BMC Medical Research Methodology*, *7*(1), 34. https://doi.org/10.1186/1471-2288-7-34

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

Murray, J. S. (2018). Multiple imputation: A review of practical and theoretical findings. *Statistical Science*, *33*(2). https://doi.org/10.1214/18-STS644

Muthén, B., & Asparouhov, T. (2012). Bayesian structural equation modeling: A more flexible representation of substantive theory. *Psychological Methods*, *17*(3), 313–335. https://doi.org/10.1037/a0026802

Nicholson, J. S., Deboeck, P. R., & Howard, W. (2017). Attrition in developmental psychology: A review of modern missing data reporting and practices. *International Journal of Behavioral Development*, *41*(1), 143–153. https://doi.org/10.1177/0165025415618275

Pandey, S., & Bright, C. L. (2008). What are degrees of freedom? *Social Work Research*, *32*(2), 119–128. https://doi.org/10.1093/swr/32.2.119

Papaioannou, A., Marsh, H. W., & Theodorakis, Y. (2004). A multilevel approach to motivational climate in physical education and sport settings: An individual or a group level construct? *Journal of Sport and Exercise Psychology*, *26*(1), 90–118. https://doi.org/10.1123/jsep.26.1.90

Peugh, J. L., & Enders, C. K. (2004). Missing data in educational research: A review of reporting practices and suggestions for improvement. *Review of Educational Research*, *74*(4), 525–556. https://doi.org/10.3102/00346543074004525

Quartagno, M., & Carpenter, J. (2023). *jomo: A package for multilevel joint modelling multiple imputation*. https://doi.org/https://CRAN.R-project.org/package=jomo

R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing.

Raudenbush, S., & Bryk, A. S. (2010). *Hierarchical linear models: Applications and data analysis methods* (2. ed.). Sage Publications, Inc.

Robitzsch, A., & Luedtke, O. (2024). *Mdmb: Model based treatment of missing data*. Comprehensive R Archive Network. https://doi.org/10.32614/CRAN.package.mdmb

Rubin, D. B. (1976). Inference and missing data. *Biometrika*, *63*(3), 581–592. https://doi.org/10.1093/biomet/63.3.581

Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. Wiley. https://doi.org/10.1002/9780470316696

Schaalje, G. B., McBride, J. B., & Fellingham, G. W. (2002). Adequacy of approximations to distributions of test statistics in complex mixed linear models. *Journal of Agricultural, Biological, and Environmental Statistics*, *7*(4), 512–524. https://doi.org/10.1198/108571102726

Schafer, J. L. (2003). Multiple imputation in multivariate problems when the imputation and analysis models differ. *Statistica Neerlandica*, *57*(1), 19–35. https://doi.org/10.1111/1467-9574.00218

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, *7*(2), 147–177. https://doi.org/10.1037/1082-989X.7.2.147

Seaman, S. R., Bartlett, J. W., & White, I. R. (2012). Multiple imputation of missing covariates with non-linear effects and interactions: An evaluation of statistical methods. *BMC Medical Research Methodology*, *12*(1), 46. https://doi.org/10.1186/1471-2288-12-46

Shin, D., & Shin, Y. (2025). *Compatible imputation for hierarchical linear models with incomplete data: Interaction effects of continuous and categorical covariates MAR* (arXiv:2502.07033). arXiv. https://doi.org/10.48550/arXiv.2502.07033

Shin, D., Shin, Y., & Hagiwara, N. (2025). Bayesian estimation of hierarchical linear models From incomplete data: Cluster-level interaction effects and small sample sizes. *Statistics in Medicine*, *44*(10-12), e70051. https://doi.org/10.1002/sim.70051

Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695

Smid, S. C., McNeish, D., Miočević, M., & Van De Schoot, R. (2020). Bayesian versus frequentist estimation for structural equation models in small sample contexts: A systematic review. *Structural Equation Modeling: A Multidisciplinary Journal*, *27*(1), 131–161. https://doi.org/10.1080/10705511.2019.1577140

Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed). Sage.

Van Buuren, S. (2018). *Flexible imputation of missing data, second edition* (2nd ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9780429492259

Van De Schoot, R., Kaplan, D., Denissen, J., Asendorpf, J. B., Neyer, F. J., & Van Aken, M. A. G. (2014). A gentle introduction to Bayesian analysis: Applications to developmental research. *Child Development*, *85*(3), 842–860. https://doi.org/10.1111/cdev.12169

Van Erp, S., Mulder, J., & Oberski, D. L. (2018). Prior sensitivity analysis in default Bayesian structural equation modeling. *Psychological Methods*, *23*(2), 363–388. https://doi.org/10.1037/met0000162

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, *16*(2). https://doi.org/10.1214/20-BA1221

Von Hippel, P. T. (2013). The bias and efficiency of incomplete-data estimators in small univariate normal samples. *Sociological Methods & Research*, *42*(4), 531–558. https://doi.org/10.1177/0049124113494582

White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, *30*(4), 377–399. https://doi.org/10.1002/sim.4067

Yuan, K.-H., & Savalei, V. (2014). Consistency, bias and efficiency of the normal-distribution-based MLE: The role of auxiliary variables. *Journal of Multivariate Analysis*, *124*, 353–370. https://doi.org/10.1016/j.jmva.2013.11.006

Yucel, R. M. (2008). Multiple imputation inference for multivariate multilevel continuous data with ignorable non-response. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *366*(1874), 2389–2403. https://doi.org/10.1098/rsta.2008.0038

Zhang, Q., & Wang, L. (2017). Moderation analysis with missing data in the predictors. *Psychological Methods*, *22*(4), 649–666. https://doi.org/10.1037/met0000104

Zitzmann, S., Lüdtke, O., & Robitzsch, A. (2015). A Bayesian approach to more stable estimates of group-level effects in contextual studies. *Multivariate Behavioral Research*, *50*(6), 688–705. https://doi.org/10.1080/00273171.2015.1090899

Appendix A

Derivations of equations of data-generating model

A.1 Calculation of σ_w^2

The data-generating model for W_j is:

$$W_j = \; a \cdot \; \bar{X}_j \; + e_{Wj}, \qquad e_{Wj} \;\; \sim \; N(0, \; {\sigma_w}^2) \tag{A.1}$$

The variance of W_j is therefore:

$$Var(W_j) = Var(a \cdot \bar{X}_j + e_{Wj}) = a^2 \cdot Var(\bar{X}_j) + Var(e_{Wj}) \tag{A.2}$$

Substituting the formula for slope calculation $a = \frac{Cov(W_j, \; \bar{X}_j)}{Var(\bar{X}_j)}$ into Equation A.2 yields:

$$\begin{aligned} Var(W_j) &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)^2} \cdot Var(\bar{X}_j) + Var(e_{Wj}) \\ &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \end{aligned}$$

This term can be simplified using $Cov(A, B) = Corr(A, B) \cdot SD(A) \cdot SD(B)$:

$$\begin{aligned} Var(W_j) &= \frac{Cov(W_j, \bar{X}_j)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= \frac{\left(Corr(W_j, \bar{X}_j) \cdot SD(W_j) \cdot SD(\bar{X}_j)\right)^2}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= \frac{Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) \cdot Var(\bar{X}_j)}{Var(\bar{X}_j)} + Var(e_{Wj}) \\ &= Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) + Var(e_{Wj}) \end{aligned}$$

Thus,

$$\begin{aligned} Var(W_j) &= Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) + Var(e_{Wj}) \\ Var(e_{Wj}) &= Var(W_j) - Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) \\ &= Var(W_j) \left(1 - Corr(W_j, \bar{X}_j)^2\right) \end{aligned}$$

Since $Var(W_j) = 1$:

$$Var(e_{Wj}) = 1 - Corr(W_j, \bar{X}_j)^2$$

In short notation:

$$\sigma_w^2 = 1 - \rho_{\bar{X}W}^2$$

A.2 Calculation of the slope to generate W_j

$$a = \frac{Cov(W_j, \bar{X}_j)}{Var(\bar{X}_j)}$$

The variance of $\bar{X}_j$ can be derived as follows:

$$\begin{aligned}
Var(\bar{X}_j) &= Var\left(\frac{1}{n}\sum_{i=1}^{n} X_{ij}\right) \\
&= Var\left(\frac{1}{n}\sum_{i=1}^{n}(u_{Xj} + e_{Xij})\right) \\
&= Var\left(\frac{1}{n}\sum_{i=1}^{n} u_{Xj} + \frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var\left(u_{Xj} + \frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + Var\left(\frac{1}{n}\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2} Var\left(\sum_{i=1}^{n} e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2}\sum_{i=1}^{n} Var(e_{Xij}) \\
&= \tau_x^2 + \frac{n \cdot \sigma_X^2}{n^2} \\
&= \tau_x^2 + \frac{\sigma_X^2}{n}
\end{aligned} \tag{A.3}$$

where n denotes the cluster size, assumed to be constant across groups. The covariance $Cov(W_j, \bar{X}_j)$ is therefore:

$$\begin{aligned}
Cov(W_j, \bar{X}_j) &= \rho_{\bar{X}W} \cdot SD(\bar{X}_j) \cdot SD(W_j) \\
&= \rho_{\bar{X}W} \cdot \sqrt{Var(\bar{X}_j)} \cdot \sqrt{Var(W_j)} \\
&= \rho_{\bar{X}W} \cdot \sqrt{\tau_x^2 + \frac{\sigma_X^2}{n}} \cdot 1
\end{aligned}$$

A.3 Variances of random components of Y

Individual and group residual variances of Y ($Var\left(u_{0j}\right) = \psi^2$ and $Var(e_{ij}) = \sigma^2$) are derived to match a prespecified residual ICC of Y:

$$ICC_{Y|X,W} = \frac{\psi^2}{\psi^2 + \sigma^2}$$

Because $Var(Y) = 1$ and the random components are uncorrelated with the predictors,

$$\psi^2 + \sigma^2 = Var(\text{Residual}) = 1 - Var(\text{fixed})$$

where $Var(\text{fixed})$ denotes the variance explained by the fixed effects. The variance components can therefore be expressed as:

$$Var(u_{0j}) = ICC_{Y|X,W} \cdot (1 - Var(\text{fixed}))$$
$$Var(e_{ij}) = (1 - ICC_{Y|X,W}) \cdot (1 - Var(\text{fixed}))$$

$Var(\text{fixed})$ can be calculated as:

$$Var(\text{fixed}) = Var(\gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + \gamma_{02}W_j)$$

The terms $(X_{ij} - \bar{X}_j)$ and $\bar{X}_j$ are uncorrelated. W_j is correlated with $\bar{X}_j$ but not with X_{ij} (see Equation Equation A.1), and $Var(W_j)$ is set to 1, which yields:

$$Var(\text{fixed}) = \gamma_{10}^2 \cdot Var(X_{ij} - \bar{X}_j) + \gamma_{01}^2 \cdot Var(\bar{X}_j) + \gamma_{02}^2 + 2\gamma_{01}\gamma_{02}Cov(\bar{X}_j, W_j)$$

Because γ_{02} is set to zero in the present design, this term can be simplified to:

$$Var(\text{fixed}) = \gamma_{10}^2 \cdot Var(X_{ij} - \bar{X}_j) + \gamma_{01}^2 \cdot Var(\bar{X}_j) \tag{A.4}$$

The variance of $(X_{ij} - \bar{X}_j)$ can be derived as follows:

$$\text{Var}(X_{ij} - \bar{X}_j) = \text{Var}(X_{ij}) + \text{Var}(\bar{X}_j) - 2 \cdot \text{Cov}(X_{ij}, \bar{X}_j)$$

$\text{Cov}(X_{ij}, \bar{X}_j)$ can be expressed as

$$\text{Cov}(X_{ij}, \bar{X}_{.j}) = \text{Cov}\left(X_{ij}, \frac{1}{n}\sum_{i'=1}^{n} X_{i'j}\right)$$

where i' may be the same or a different index than i. Using the decomposition of X_{ij} introduced in Section Section 3.1.1, $X_{ij} = u_{Xj} + e_{Xij}$, the covariance term can be rewritten and expanded as

$$\begin{aligned}
\text{Cov}(X_{ij}, \bar{X}_j) &= \text{Cov}\left(u_{Xj} + e_{Xij}, \frac{1}{n}\sum_{i'=1}^{n}(u_{Xj} + e_{Xi'j})\right) \\
&= \text{Cov}\left(u_{Xj} + e_{Xij},\ u_{Xj} + \frac{1}{n}\sum_{i'=1}^{n} e_{Xi'j}\right) \\
&= \text{Cov}(u_{Xj}, u_{Xj}) + \text{Cov}\left(u_{Xj}, \frac{1}{n}\sum_{i'=1}^{n} e_{Xi'j}\right) \\
&\quad + \text{Cov}(e_{Xij}, u_{Xj}) + \text{Cov}\left(e_{Xij}, \frac{1}{n}\sum_{i'=1}^{n} e_{Xi'j}\right)
\end{aligned}$$

Because of uncorrelatedness between u_{Xj} and e_{Xij}, the second and third term are zero; furthermore, $\text{Cov}\left(e_{Xij}, \frac{1}{n}\sum_{i'=1}^{n} e_{Xi'j}\right) = \frac{1}{n}\sum_{i'=1}^{n}\text{Cov}(e_{Xij}, e_{Xi'j})$.
As $\text{Cov}(e_{Xij}, e_{Xi'j}) = \text{Var}(e_{Xij})$ if $i = i'$ and zero otherwise, $\frac{1}{n}\sum_{i'=1}^{n}\text{Cov}(e_{Xij}, e_{Xi'j}) = \frac{1}{n}\text{Var}(e_{Xij})$.

Overall, the covariance now simplifies to

$$\begin{aligned}\mathrm{Cov}(X_{ij}, \bar{X}_j) &= \mathrm{Var}(u_{Xj}) + \frac{1}{n}\mathrm{Var}(e_{Xij}) \\ &= \tau_x^2 + \frac{\sigma_X^2}{n} \\ &= \mathrm{Var}(\bar{X}_j)\end{aligned}$$

Returning to $\mathrm{Var}(X_{ij} - \bar{X}_j)$, this simplifies to:

$$\begin{aligned}\mathrm{Var}(X_{ij} - \bar{X}_j) &= \mathrm{Var}(X_{ij}) + \mathrm{Var}(\bar{X}_j) - 2 \cdot \mathrm{Cov}(X_{ij}, \bar{X}_j) \\ &= \mathrm{Var}(X_{ij}) + \mathrm{Var}(\bar{X}_j) - 2 \cdot \mathrm{Var}(\bar{X}_j) \\ &= \mathrm{Var}(X_{ij}) - \mathrm{Var}(\bar{X}_j) \\ &= (\tau_x^2 + \sigma_X^2) - (\tau_x^2 + \frac{\sigma_X^2}{n}) \\ &= \sigma_X^2 - \frac{\sigma_X^2}{n}\end{aligned}$$

Inserting the derived terms into Equation A.4, $Var(\text{fixed})$ can be therefore calculated as

$$Var(\text{fixed}) = \gamma_{10}^2 \left(\sigma_X^2 - \frac{\sigma_X^2}{n}\right) + \gamma_{01}^2 \left(\tau_x^2 + \frac{\sigma_X^2}{n}\right).$$

The variances of the random components can thus be calculated like this:

$$\psi^2 = ICC_{Y|X,W} \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n}))$$

$$\sigma^2 = (1 - ICC_{Y|X,W}) \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n}))$$

Appendix B

Convergence analysis for MI and fully Bayesian MCMC

Convergence behavior was tested in one run of the most challenging condition ($N_2 = 15, \gamma_{01} = 0.4, \beta = .3, ICC = .1$).

B.1 Diagnostics of multiple imputation

Convergence of the imputation model was assessed using potential scale reduction factors (Gelman & Rubin, 1992). Scale reduction factors that are substantially larger than 1 indicate that convergence was not yet reached fully; an acceptable threshold can be set at 1.1 (Gelman et al., 2025, pp. 285–287), while a stricter recommendation was recently set to be 1.01 (Vehtari et al., 2021). Given the simulation setting with aggregated results over many replications, the first threshold was considered sufficient.

All values were close to 1, indicating satisfactory convergence across all parameters. A maximum $\hat{R}$ of 1.026 was observed for the intercept of X in the imputation model. Diagnostic plots are shown in Figure B.1 - B.3 for this parameter and the most critical parameters, the level-2 variances of Y and X.

Trace plots showed no signs of substantial change after burn-in. Inspection of the autocorrelation plots indicated that autocorrelation diminished quickly across increasing lags and did not persist over longer distances between samples. The distance of 300 lags between imputation that was chosen in this simulation was therefore sufficient.

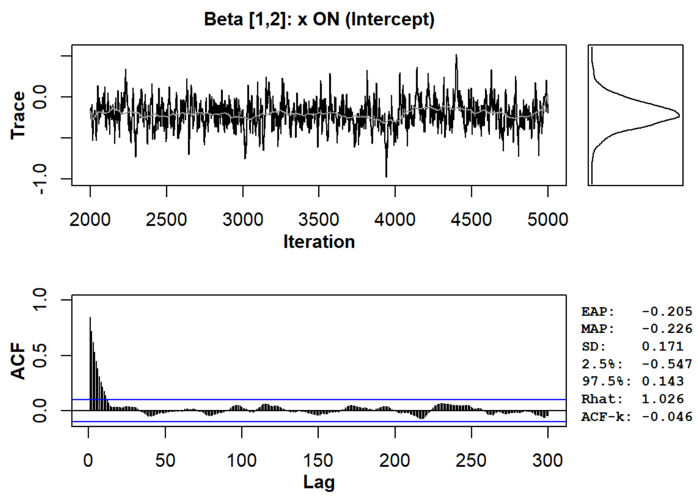

FIGURE B.1: Trace and autocorrelation plot in an exemplary simulation run for the intercept in the imputation model for X (multiple imputation).

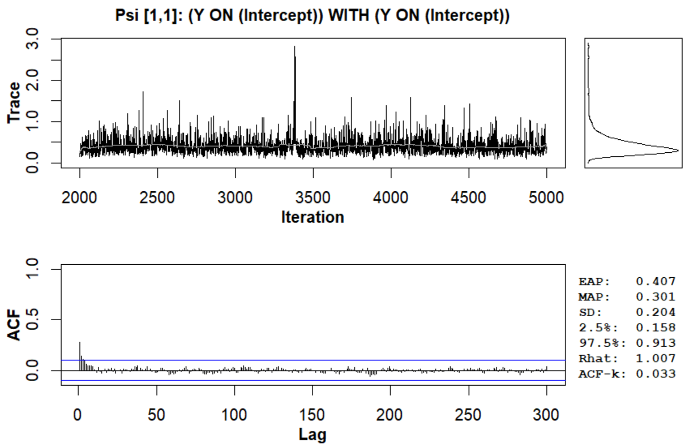

FIGURE B.2: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of Y (multiple imputation).

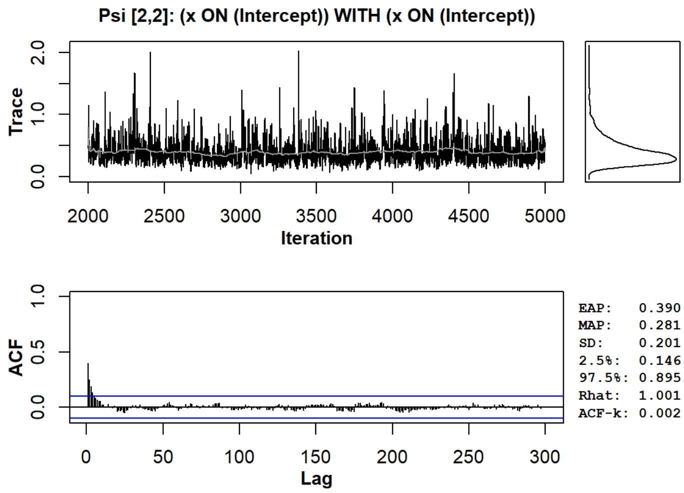

FIGURE B.3: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of X (multiple imputation).

B.2 Diagnostics of fully Bayesian analysis

$\hat{R}$ values were close to 1, except for the level-2 variance of X ($\hat{R} = 1.05$) and the regression coefficient of the group mean of X on W ($\hat{R} = 1.03$). Diagnostic plots of these parameters and the additionally critical parameter τ_{y0}^2 are shown in Figure B.4. Trace plots showed no drifting. Autocorrelation decreased rapidly for most parameters, although some exhibited persistent but low autocorrelation across lags. This pattern suggests slightly reduced mixing efficiency rather than substantial convergence problems (Johnson et al., 2022, pp. 149–152).

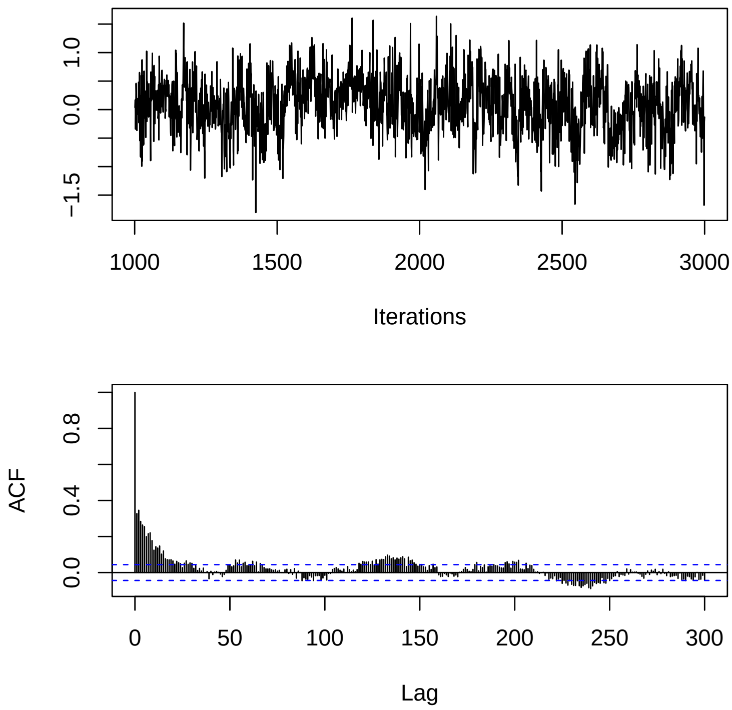

FIGURE B.4: Trace and autocorrelation plot for the effect of the group mean of X on W in the imputation model (fully Bayesian estimation).

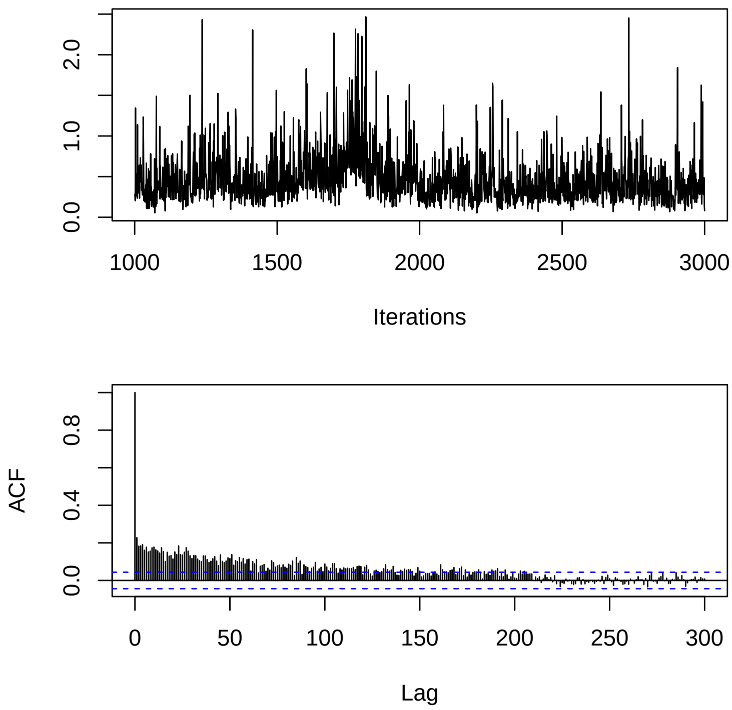

FIGURE B.5: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of X (fully Bayesian analysis)

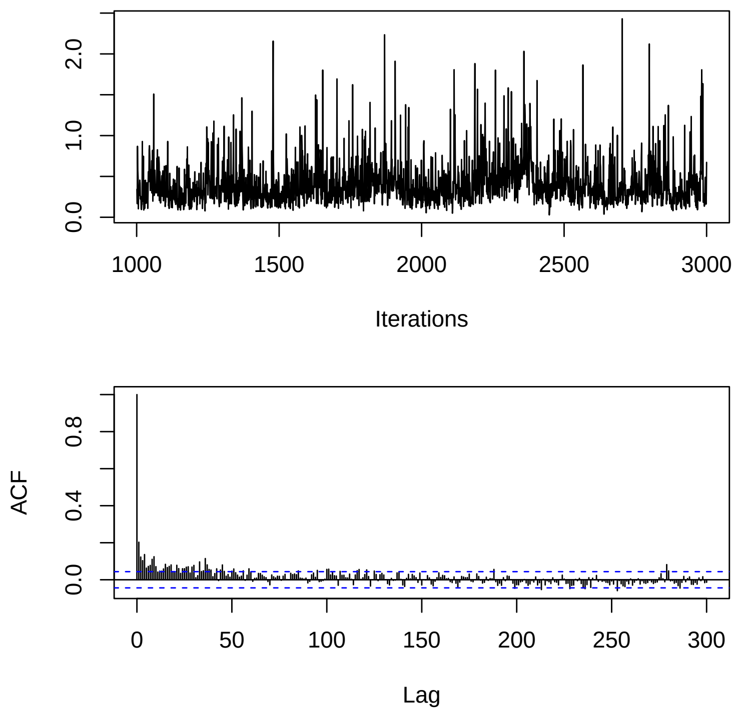

FIGURE B.6: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of Y (fully Bayesian analysis)

Appendix C

Complete simulation results

Table C.1: Simulation results ($N_2 = 15$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
Parameter	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.002	0.942	0.284	1.170	0.037	0.941	0.300	1.180	0.005	0.966	0.252	1.115	0.005	0.986	0.252	1.284	0.003	0.980	0.291	1.456
γ_{10}	-0.006	0.951	0.084	0.335	-0.006	0.944	0.104	0.414	-0.008	0.943	0.103	0.409	-0.008	0.944	0.103	0.414	-0.010	0.946	0.100	0.401
ICC = 0.3																				
γ_{01}	0.000	0.960	0.272	1.174	0.012	0.953	0.280	1.177	-0.001	0.942	0.278	1.126	-0.001	0.972	0.278	1.278	-0.003	0.980	0.288	1.384
γ_{10}	0.001	0.942	0.089	0.336	0.000	0.931	0.111	0.415	-0.005	0.938	0.108	0.411	-0.005	0.939	0.108	0.416	-0.006	0.934	0.108	0.405
MAR																				
ICC = 0.1																				
γ_{01}	-0.002	0.949	0.283	1.192	0.031	0.947	0.284	1.189	0.007	0.970	0.247	1.131	0.007	0.985	0.247	1.305	0.001	0.987	0.289	1.481
γ_{10}	0.000	0.942	0.087	0.336	-0.002	0.954	0.106	0.416	-0.007	0.949	0.104	0.412	-0.007	0.950	0.104	0.418	-0.007	0.954	0.102	0.403
ICC = 0.3																				
γ_{01}	-0.016	0.943	0.279	1.175	0.007	0.949	0.279	1.175	-0.014	0.943	0.275	1.130	-0.014	0.968	0.275	1.285	-0.016	0.975	0.291	1.390
γ_{10}	0.001	0.941	0.087	0.337	-0.001	0.948	0.105	0.414	-0.004	0.943	0.102	0.411	-0.004	0.945	0.102	0.417	-0.005	0.949	0.101	0.405

Note:
Bias = average deviation from the true parameter; Cov = empirical coverage rate; SD = empirical standard deviation; CIW = mean confidence/credible interval width. MCAR = missing completely at random; MAR = missing at random. CD = complete data; LD = listwise deletion; MI-R = multiple imputation using Rubin's degrees-of-freedom approximation; MI-a = multiple imputation using small-sample-adjusted degrees of freedom; Bayes = fully Bayesian estimation.

Table C.2: Simulation results ($N_2 = 15$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	-0.008	0.953	0.275	1.174	-0.038	0.968	0.269	1.183	-0.075	0.965	0.239	1.151	-0.075	0.981	0.239	1.329	-0.036	0.985	0.281	1.516
γ_{10}	0.000	0.952	0.083	0.330	0.000	0.952	0.102	0.404	-0.001	0.950	0.101	0.401	-0.001	0.953	0.101	0.407	-0.003	0.954	0.099	0.393
ICC = 0.3																				
γ_{01}	0.003	0.955	0.272	1.154	-0.013	0.949	0.272	1.156	-0.016	0.951	0.275	1.122	-0.016	0.967	0.275	1.277	-0.008	0.971	0.290	1.385
γ_{10}	0.002	0.949	0.083	0.326	0.003	0.935	0.106	0.402	-0.001	0.944	0.103	0.397	-0.001	0.948	0.103	0.402	-0.003	0.942	0.101	0.391
MAR																				
ICC = 0.1																				
γ_{01}	-0.004	0.959	0.276	1.151	-0.050	0.942	0.283	1.169	-0.078	0.965	0.238	1.124	-0.078	0.977	0.238	1.299	-0.039	0.982	0.276	1.461
γ_{10}	-0.007	0.954	0.084	0.331	-0.005	0.952	0.105	0.408	-0.007	0.956	0.102	0.404	-0.007	0.959	0.102	0.409	-0.010	0.952	0.101	0.396
ICC = 0.3																				
γ_{01}	0.006	0.962	0.269	1.162	-0.019	0.956	0.275	1.165	-0.020	0.947	0.271	1.137	-0.020	0.976	0.271	1.296	-0.008	0.977	0.290	1.407
γ_{10}	-0.001	0.945	0.086	0.327	0.000	0.948	0.105	0.403	-0.003	0.946	0.102	0.398	-0.003	0.948	0.102	0.403	-0.006	0.951	0.102	0.393

Note:
Abbreviations as listed in Table C.1

TABLE C.3: Simulation results ($N_2 = 30$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.003	0.945	0.188	0.755	0.042	0.939	0.186	0.757	0.014	0.962	0.173	0.766	0.014	0.976	0.173	0.812	0.009	0.963	0.199	0.864
γ_{10}	-0.005	0.953	0.059	0.236	-0.006	0.953	0.073	0.291	-0.007	0.954	0.073	0.290	-0.007	0.954	0.073	0.292	-0.008	0.950	0.072	0.283
ICC = 0.3																				
γ_{01}	-0.004	0.957	0.185	0.758	0.010	0.964	0.185	0.760	-0.003	0.955	0.190	0.769	-0.003	0.964	0.190	0.810	-0.003	0.965	0.192	0.826
γ_{10}	0.001	0.942	0.063	0.237	0.000	0.943	0.077	0.291	-0.003	0.939	0.077	0.291	-0.003	0.940	0.077	0.293	-0.003	0.943	0.076	0.284
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.941	0.191	0.757	0.039	0.947	0.192	0.761	0.010	0.965	0.177	0.767	0.010	0.972	0.177	0.813	0.003	0.965	0.203	0.870
γ_{10}	0.000	0.944	0.061	0.236	-0.002	0.945	0.075	0.290	-0.003	0.950	0.074	0.288	-0.003	0.951	0.074	0.290	-0.004	0.948	0.073	0.282
ICC = 0.3																				
γ_{01}	0.008	0.969	0.179	0.765	0.027	0.956	0.183	0.768	0.011	0.958	0.186	0.777	0.011	0.968	0.186	0.820	0.010	0.967	0.188	0.836
γ_{10}	0.000	0.944	0.062	0.238	0.000	0.948	0.076	0.293	-0.003	0.954	0.075	0.294	-0.003	0.954	0.075	0.296	-0.003	0.946	0.074	0.286

Note:
Abbreviations as listed in Table C.1

Table C.4: Simulation results (N_2 = 30, γ_{01} = 0.4)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.005	0.950	0.189	0.744	-0.034	0.956	0.187	0.748	-0.052	0.956	0.175	0.767	-0.052	0.966	0.175	0.813	-0.009	0.970	0.204	0.865
γ_{10}	0.000	0.966	0.056	0.231	-0.001	0.967	0.068	0.284	0.001	0.966	0.066	0.283	0.001	0.969	0.066	0.285	-0.003	0.967	0.065	0.277
ICC = 0.3																				
γ_{01}	0.007	0.943	0.188	0.736	-0.008	0.941	0.191	0.740	-0.002	0.938	0.194	0.753	-0.002	0.953	0.194	0.794	0.004	0.951	0.201	0.811
γ_{10}	0.000	0.963	0.057	0.229	0.003	0.956	0.069	0.282	0.001	0.950	0.067	0.281	0.001	0.954	0.067	0.283	0.000	0.954	0.067	0.275
MAR																				
ICC = 0.1																				
γ_{01}	0.005	0.948	0.191	0.744	-0.033	0.944	0.190	0.749	-0.057	0.952	0.174	0.772	-0.057	0.963	0.174	0.819	-0.015	0.958	0.204	0.868
γ_{10}	0.001	0.947	0.060	0.233	0.001	0.946	0.073	0.285	0.003	0.948	0.073	0.285	0.003	0.950	0.073	0.287	0.000	0.946	0.071	0.277
ICC = 0.3																				
γ_{01}	0.002	0.941	0.182	0.737	-0.014	0.947	0.180	0.738	-0.006	0.942	0.185	0.754	-0.006	0.951	0.185	0.795	0.000	0.954	0.191	0.813
γ_{10}	-0.002	0.948	0.059	0.230	-0.001	0.947	0.075	0.284	-0.003	0.950	0.072	0.283	-0.003	0.951	0.072	0.285	-0.004	0.948	0.072	0.277

Note:
Abbreviations as listed in Table C.1

Table C.5: Simulation results ($N_2 = 60$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	-0.006	0.955	0.130	0.516	0.027	0.954	0.128	0.517	-0.002	0.971	0.124	0.542	-0.002	0.979	0.124	0.556	-0.010	0.961	0.139	0.575
γ_{10}	0.000	0.944	0.043	0.167	-0.001	0.953	0.053	0.205	-0.002	0.939	0.053	0.205	-0.002	0.940	0.053	0.206	-0.002	0.945	0.052	0.199
ICC = 0.3																				
γ_{01}	-0.002	0.946	0.127	0.518	0.017	0.955	0.127	0.519	0.001	0.949	0.132	0.536	0.001	0.954	0.132	0.549	0.000	0.954	0.133	0.550
γ_{10}	0.000	0.947	0.042	0.167	0.002	0.947	0.052	0.206	0.000	0.948	0.051	0.206	0.000	0.948	0.051	0.206	0.000	0.946	0.051	0.201
MAR																				
ICC = 0.1																				
γ_{01}	0.000	0.951	0.131	0.517	0.037	0.949	0.129	0.520	0.008	0.960	0.126	0.545	0.008	0.965	0.126	0.559	0.001	0.953	0.142	0.582
γ_{10}	-0.001	0.954	0.043	0.167	0.000	0.954	0.052	0.204	-0.001	0.956	0.051	0.205	-0.001	0.957	0.051	0.205	-0.001	0.953	0.050	0.198
ICC = 0.3																				
γ_{01}	0.005	0.948	0.131	0.522	0.021	0.949	0.131	0.522	0.003	0.946	0.137	0.539	0.003	0.949	0.137	0.552	0.003	0.956	0.137	0.556
γ_{10}	0.001	0.954	0.041	0.167	0.000	0.949	0.052	0.205	-0.001	0.947	0.052	0.205	-0.001	0.947	0.052	0.206	-0.001	0.944	0.051	0.201

Note:
Abbreviations as listed in Table C.1

TABLE C.6: Simulation results ($N_2 = 60$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.001	0.946	0.126	0.509	-0.039	0.939	0.124	0.509	-0.044	0.955	0.121	0.541	-0.044	0.960	0.121	0.556	-0.008	0.952	0.136	0.577
γ_{10}	0.001	0.960	0.041	0.164	0.001	0.961	0.050	0.201	0.004	0.954	0.049	0.200	0.004	0.955	0.049	0.200	0.001	0.958	0.049	0.195
ICC = 0.3																				
γ_{01}	-0.002	0.952	0.129	0.501	-0.021	0.947	0.128	0.502	-0.007	0.954	0.132	0.521	-0.007	0.957	0.132	0.534	-0.005	0.962	0.133	0.536
γ_{10}	-0.001	0.954	0.040	0.162	-0.001	0.956	0.049	0.199	-0.002	0.956	0.048	0.199	-0.002	0.956	0.048	0.200	-0.002	0.961	0.048	0.194
MAR																				
ICC = 0.1																				
γ_{01}	0.000	0.954	0.129	0.509	-0.035	0.943	0.128	0.512	-0.044	0.960	0.125	0.544	-0.044	0.962	0.125	0.559	-0.008	0.957	0.144	0.580
γ_{10}	0.002	0.926	0.043	0.163	0.001	0.942	0.051	0.200	0.003	0.941	0.050	0.199	0.003	0.942	0.050	0.200	0.001	0.941	0.050	0.194
ICC = 0.3																				
γ_{01}	-0.005	0.950	0.127	0.503	-0.022	0.956	0.125	0.504	-0.009	0.958	0.130	0.523	-0.009	0.961	0.130	0.536	-0.008	0.962	0.132	0.540
γ_{10}	-0.001	0.953	0.042	0.162	-0.002	0.944	0.051	0.199	-0.003	0.944	0.050	0.199	-0.003	0.945	0.050	0.200	-0.003	0.942	0.050	0.195

Note:
Abbreviations as listed in Table C.1

Table C.7: MCSEs of performance measures ($N_2 = 15$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.010	0.009	0.007	0.007	0.010	0.008	0.006	0.006	0.010	0.008	0.004	0.006	0.012	0.009	0.004	0.007	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.010	0.009	0.007	0.006	0.011	0.009	0.005	0.006	0.012	0.009	0.004	0.006	0.013
γ_{10}	0.003	0.007	0.002	0.001	0.004	0.008	0.002	0.001	0.003	0.008	0.002	0.002	0.003	0.008	0.002	0.002	0.003	0.008	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.008	0.005	0.006	0.011	0.008	0.004	0.006	0.013	0.009	0.004	0.006	0.016
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.006	0.006	0.013	0.009	0.005	0.007	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

Table C.8: MCSEs of performance measures ($N_2 = 15$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.010	0.008	0.006	0.006	0.010	0.008	0.006	0.005	0.010	0.008	0.004	0.005	0.013	0.009	0.004	0.006	0.020
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.006	0.006	0.013	0.009	0.005	0.006	0.016
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.008	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.010	0.008	0.006	0.005	0.010	0.008	0.005	0.005	0.012	0.009	0.004	0.006	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.006	0.002	0.002	0.003	0.006	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.008	0.006	0.006	0.010	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.011	0.009	0.005	0.006	0.013	0.009	0.005	0.006	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.9: MCSEs of performance measures ($N_2 = 30$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.008	0.004	0.004	0.005	0.006	0.004	0.005	0.005	0.005	0.004	0.005	0.006	0.006	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.004	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.005
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.008	0.002	0.001	0.002	0.008	0.002	0.001	0.002	0.007	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.005	0.004	0.006	0.006	0.006	0.005	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.005	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.006	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.10: MCSEs of performance measures ($N_2 = 30$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.004	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.005	0.005	0.006
γ_{10}	0.002	0.006	0.001	0.000	0.002	0.006	0.002	0.001	0.002	0.006	0.001	0.001	0.002	0.005	0.001	0.001	0.002	0.006	0.001	0.001
ICC = 0.3																				
γ_{01}	0.006	0.007	0.004	0.004	0.006	0.007	0.004	0.004	0.006	0.008	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.005	0.005
γ_{10}	0.002	0.006	0.001	0.000	0.002	0.006	0.002	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.004	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.006	0.005	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.006	0.006	0.007	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

Table C.11: MCSEs of performance measures ($N_2 = 60$, $\gamma_{01} = 0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.005	0.003	0.003	0.004	0.005	0.003	0.003	0.004	0.006	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.008	0.001	0.001	0.002	0.008	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.007	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000
MAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.003	0.004	0.007	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.006	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.12: MCSEs of performance measures ($N_2 = 60$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.008	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.006	0.003	0.003	0.004	0.007	0.003	0.003
γ_{10}	0.001	0.006	0.001	0.000	0.002	0.006	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.006	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.006	0.001	0.000	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.000
MAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.003	0.005	0.006	0.003	0.003
γ_{10}	0.001	0.008	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000

Note:
All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

Appendix D

Default priors and sensitivity analysis

D.1 Default priors in mdmb

Default priors in mdmb are noninformative. For regression coefficients, a uniform prior on the interval $(-\infty, \infty)$ is approximated by a normal distribution with large variance, i.e., $N(0, 10^{10})$. For variance–covariance matrices on level 1 and 2, inverse-Wishart priors are used:

$$\Sigma \sim IW(\nu, \mathbf{S})$$

Here, Σ denotes a generic covariance matrix (σ at level 1 and τ at level 2), ν denotes the degrees of freedom, and $\mathbf{S}$ is the scale matrix. The default in mdmb is set to improper inverse-Wishart priors with negative degrees of freedom, which can be written as $IW(-p-1, \mathbf{0})$ (Asparouhov & Muthen, 2010), where p denotes the dimension of the covariance matrix. $\mathbf{S} = \mathbf{0}$ corresponds to a null matrix of appropriate dimension.

D.2 Default priors in jomo

In jomo, inverse-Wishart priors are used for the covariance matrices at level 1 and 2 for imputation. The default in the mitml implementation is set to “least informative” proper priors with largest dispersion, i.e., smallest possible degrees of freedom, scaled by the identity matrix (Grund et al., 2015; Quartagno & Carpenter, 2023). For the three variables of the imputation model (X, Y, Z) in this thesis the scale matrix is:

$$\mathbf{S} = \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

D.3 Sensitivity analysis with weakly informative priors in jomo

In this analysis, for the conditions when multiple imputation produced the highest bias, i.e. when $ICC = .1$ and $\gamma_{01} = 0.4$, alternative inverse Wishart priors were specified for the level-2 covariance matrix. This was done by specifying a plausible prior guess for the level-2 covariance matrix. In the simulation study, this guess was based on the true parameter values; in empirical applications, it could be derived from previous research. This guess for the covariance matrix corresponds to the two standardized

TABLE D.1: Bias reduction with alternative priors

N2	**Bias.d**	**rel. Bias.d**	**Bias.a**	**rel. Bias.a**	**Bias red.**
MCAR					
15	-0.075	18.7%	-0.048	12.1%	35.5%
30	-0.062	15.4%	-0.033	8.3%	46.2%
60	-0.040	9.9%	-0.014	3.6%	63.7%
MAR					
15	-0.068	17.1%	-0.037	9.1%	46.6%
30	-0.059	14.8%	-0.033	8.2%	44.3%
60	-0.045	11.2%	-0.021	5.3%	52.3%

Note:
d = default priors, a = alternative priors, Bias red. = Bias reduction

level-1 variables X and Y (ICC = .1) and the standardized level-2 variable W:

$$\tau_{\text{guess}} = \begin{bmatrix} .1 & 0 & 0 \\ 0 & .1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

τ denotes the level-2 covariance matrix of the random effects. The level-2 covariance prior was then defined as

$$\tau \sim IW(\nu, \mathbf{S}), \qquad \mathbf{S} = 3\,\tau_{\text{guess}}$$

Only the scale matrix was modified, while the degrees of freedom were kept at their default value, such that the prior remained weakly informative but was centered around more plausible variance values. The level-1 covariance matrix was kept at the default specification.

Bias was substantially reduced across all investigated conditions, with reductions ranging from approximately 35% to over 60% and increasing with larger number of groups (see Table D.1). Coverage rates remained stable, while empirical standard deviations showed a small increase under the alternative prior specification.

Affidavit

"I hereby declare under oath that I have written this thesis independently and have not used any resources, including additional internet sources, other than those specified in the bibliography. I have not previously submitted the thesis for any other examination procedure, and that the written version corresponds to the version on the digital storage medium.

I further declare that I have fully documented all computational/AI-assisted writing tools used in preparing this thesis in the "Overview of Use" section, including product names and how they were used.

Throughout the preparation and writing of this thesis, I have maintained independent authorship and when using computational/AI-supported writing tools, I have exercised direct control and oversight over their application."

Hamburg, May 5, 2026

Signature

Declaration on the Use of Generative Artificial Intelligence

In writing this thesis, I used the following generative artificial intelligence (gAI) systems:

1. ChatGPT
2. DeepL

I further declare that I

- have actively informed myself about the capabilities and limitations of the above-mentioned gAI systems,
- have marked passages taken verbatim from the above-mentioned gAI systems,
- have verified that the content generated using the above-mentioned gAI systems and adopted by me is factually correct,
- am aware that, as the author of this work, I am responsible for the information and statements made therein.

I have used the above-mentioned gAI systems for the purposes described on the next page. The specific system is not listed for each task, as ChatGPT was used across all stages, except for translation and language refinement, where DeepL was additionally used.

Tabular Representation of the Use of AI Systems

Work Step	**Description of Use**
Programming and Simulation Development	
Debugging	Debugging of R code used in simulation studies
Code understanding	Assistance in understanding and implementing simulation code
Data handling	Support in handling and processing simulation output data
Statistical and Methodological Support	
Conceptual understanding	Assistance in understanding statistical concepts and procedures
Equations	Support in transforming equations
Literature Work	
Literature research	Assistance in identifying relevant literature when needed
Literature understanding	Assistance in understanding complex passages in scientific literature
Data Visualization	
Generating visualizations	Assistance in generating and refining graphical representations of simulation results in R
Equations in diagrams	Assistance with integrating LaTeX equation rendering into Inkscape
Writing and Text Development	
Text style	Improving clarity, structure, and academic style of text passages
Translations	Translating text into English (in combination with DeepL)
Argumentation	Providing suggestions for revision and improvement
Technical Implementation of the Thesis Document	
Markdown/Quarto	Assistance with the use of Quarto and Markdown and help with debugging
Formatting	Support in formatting and adjusting the template (e.g., tables, graphics, layout adjustments, and LaTeX integration)

Hamburg, May 5, 2026

Signature